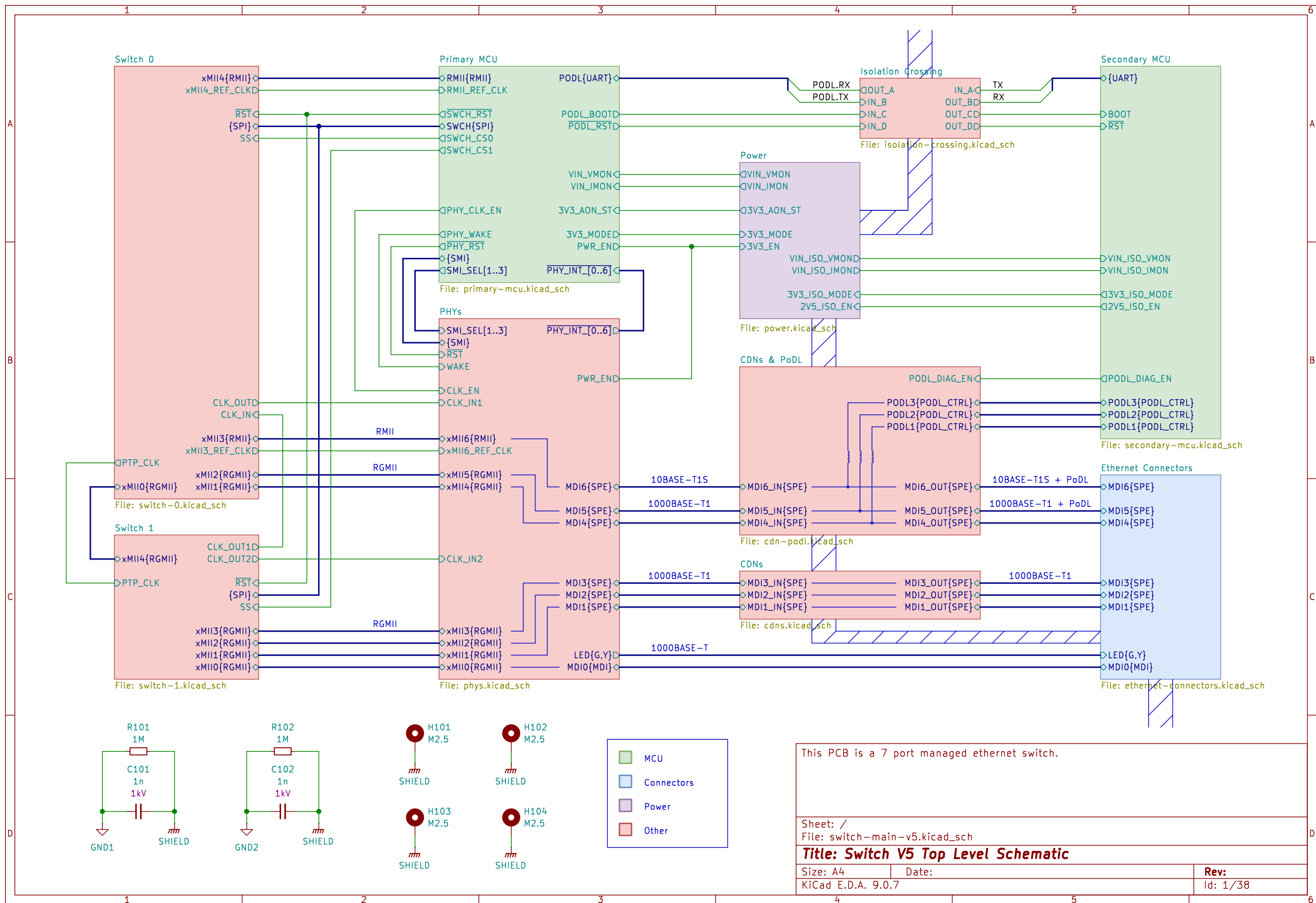
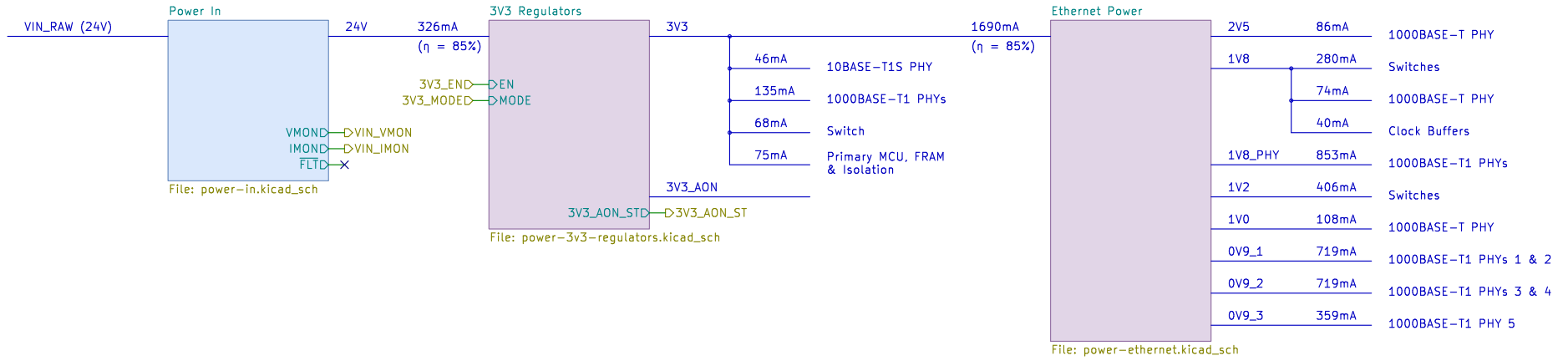


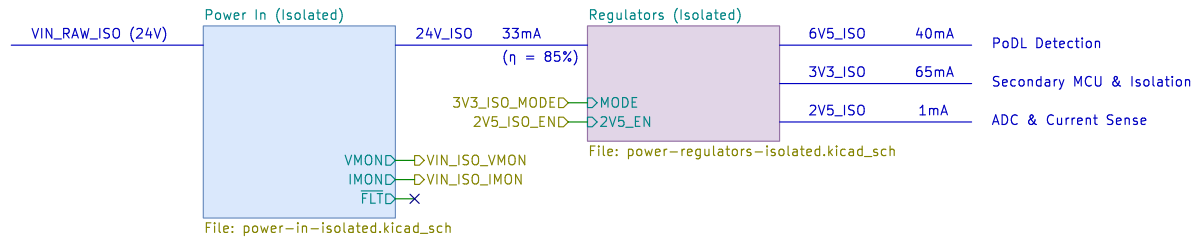


PHY Side Power



Isolation Barrier

Connector Side Power



Block diagram of the power supply scheme for both sides of the isolation barrier.

Sheet: /Power/
File: power.kicad_sch

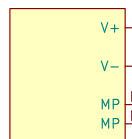
Title: Power	
Size: A4	
KiCad E.D.A. 9.0.7	

Date:

Rev:

Id: 2/38

J301
MKXS1-1 Power Input



ESD
Protection

Surge
Protection

VIN_RAW

D301
SZESD7241N2T5G



U302
TVS3301DRBR

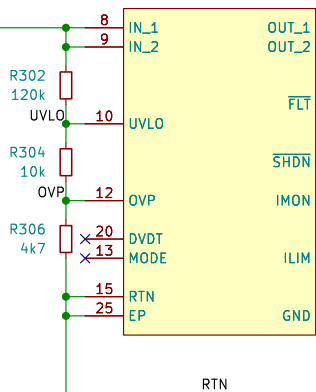
C301
100n
100V

GND1

UVLO = 10.9V
OVP = 34.1V

eFuse

U301
TPS26600RHFR



$VMON = VIN / 18.857$
The 150mΩ eFuse
resistance should be
taken into account when
calculating input power.

+VIN

R301
100k
VMON
R303
5k6
GND1

$I_{OUT} = VMON / 7.84$

$R_{ILIM} = 22k\Omega$
 $\therefore I_{LIM} = 12 / 22 = 545mA$

TPS26600 has auto-retry
when a fault occurs.

Power input and protection for the PHY side of the PCB.

Sheet: /Power/Power In/
File: power-in.kicad_sch

Title: Power In (PHY Side)

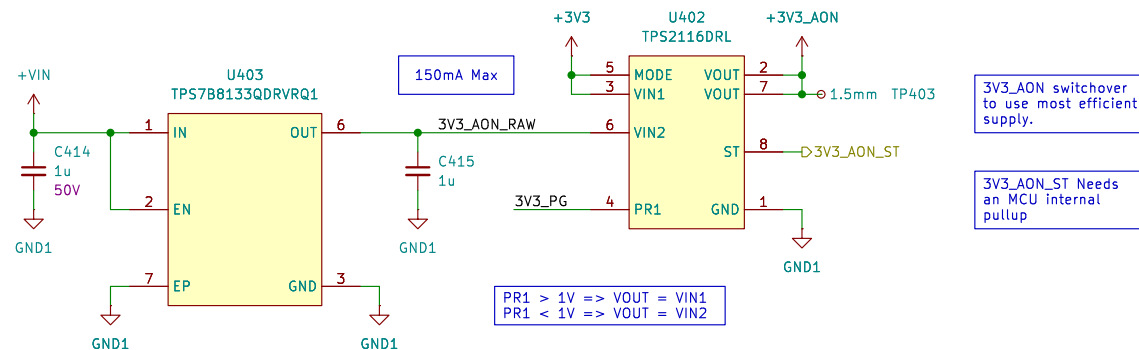
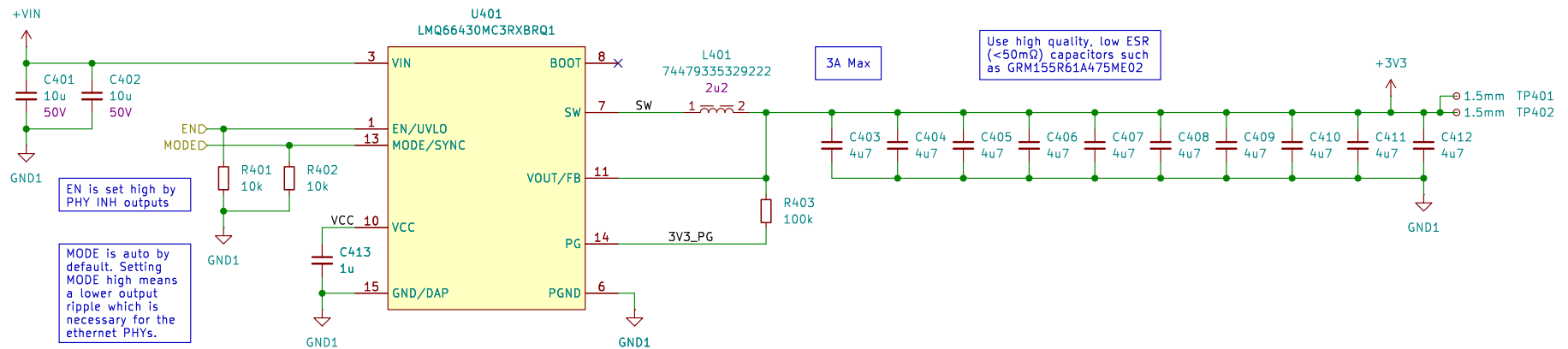
Size: A4

Date:

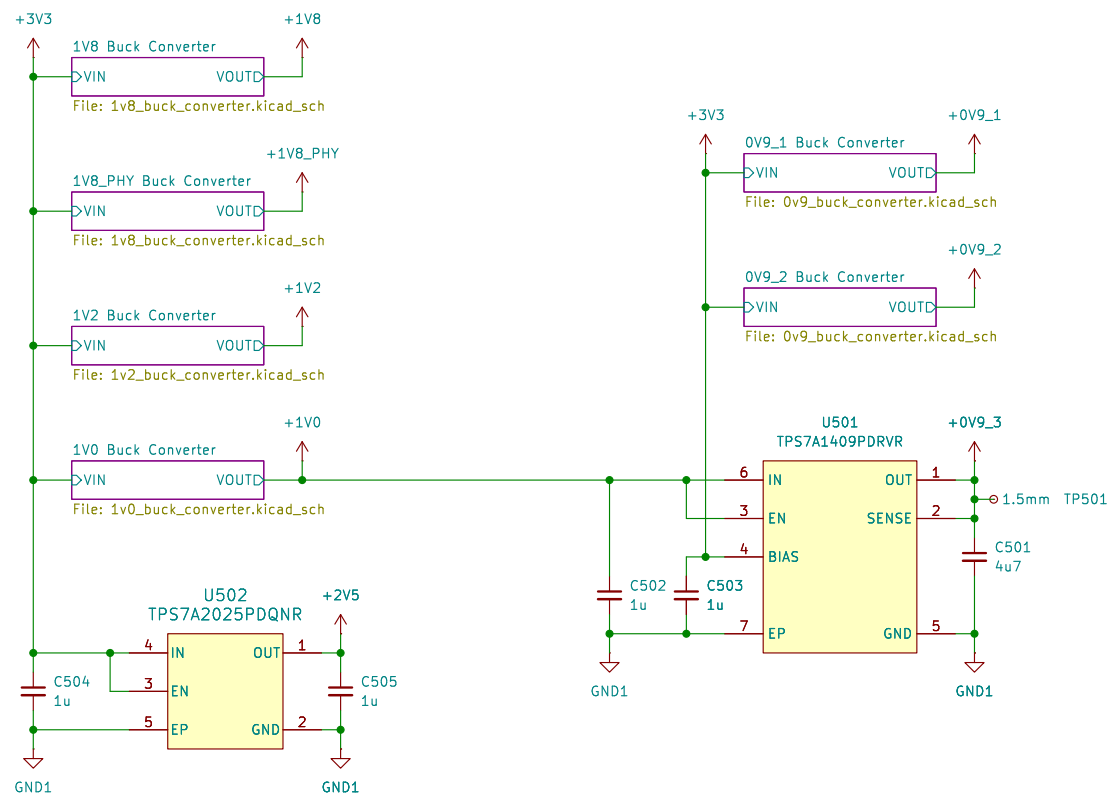
KiCad E.D.A. 9.0.7

Rev:

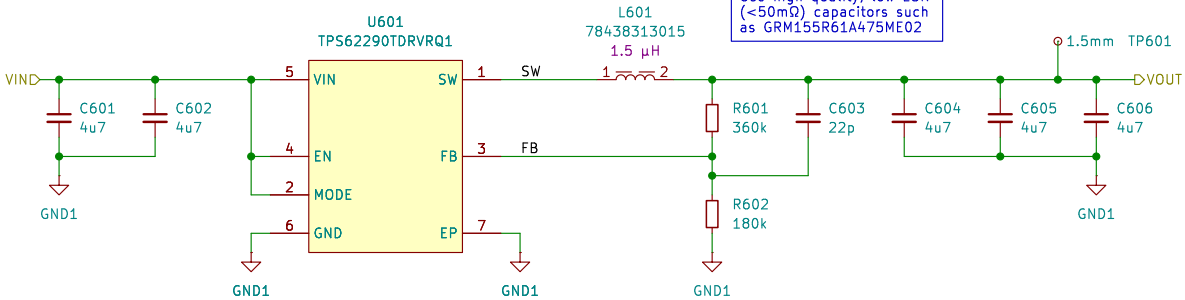
Id: 3/38



Rev:
Id: 4/38



1V8 Regulator



Use high quality, low ESR (<50mΩ) capacitors such as GRM155R61A475ME02

1.8V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V8 Buck Converter/
File: 1v8_buck_converter.kicad_sch

Title:

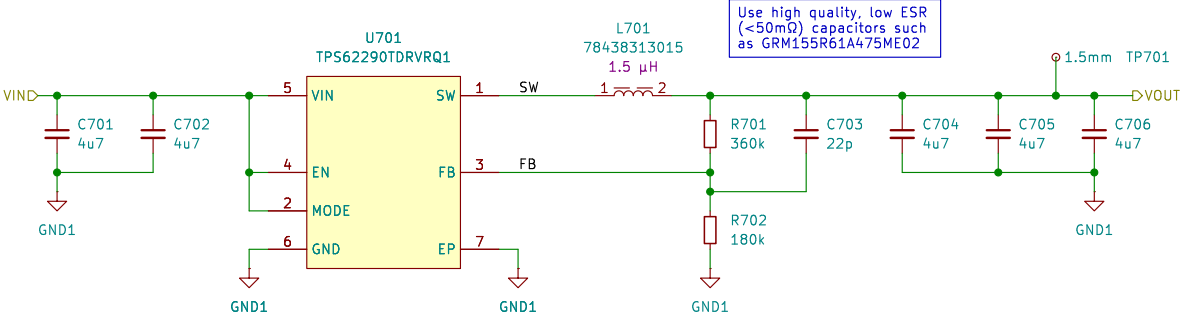
Size: A4	
KiCad E.D.A. 9.0.7	

Date:

Rev:

Id: 6/38

1V8 Regulator



1.8V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V8_PHY Buck Converter/
File: 1v8_buck_converter.kicad_sch

Title:

Size: A4

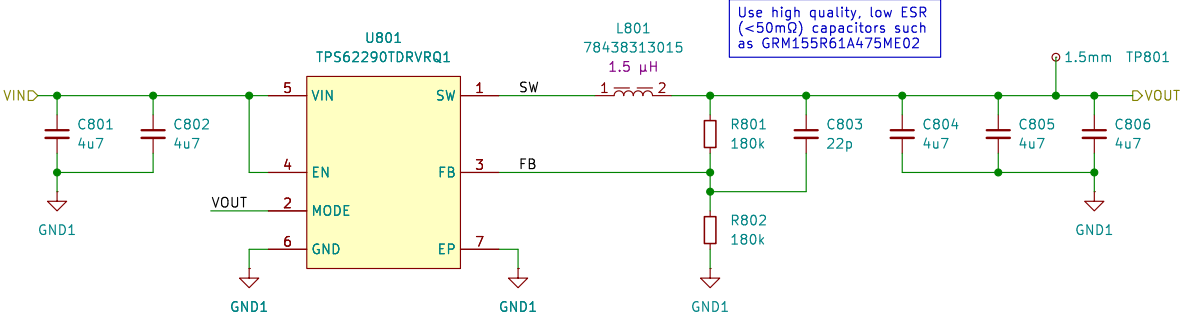
Date:

KiCad E.D.A. 9.0.7

Rev:

Id: 7/38

1V2 Regulator



1.2V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V2 Buck Converter/
File: 1v2_buck_converter.kicad_sch

Title:

Size: A4

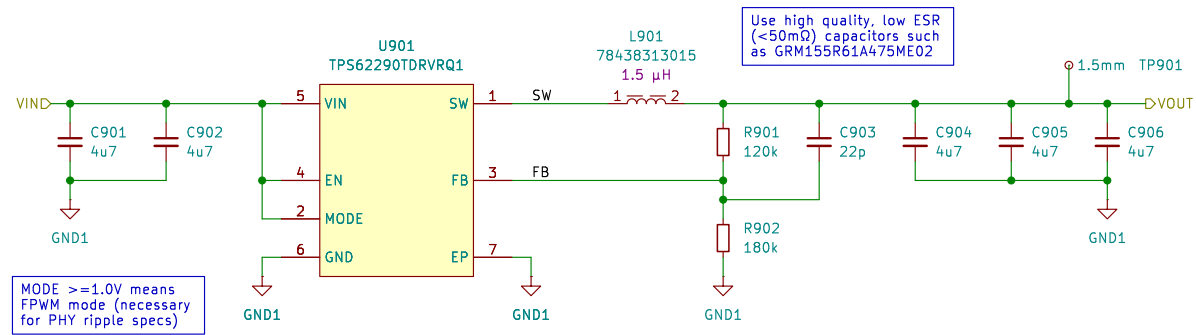
Date:

KiCad E.D.A. 9.0.7

Rev:

Id: 8/38

1V0 Regulator



1.0V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V0 Buck Converter/
File: 1v0_buck_converter.kicad_sch

Title:

Size: A4

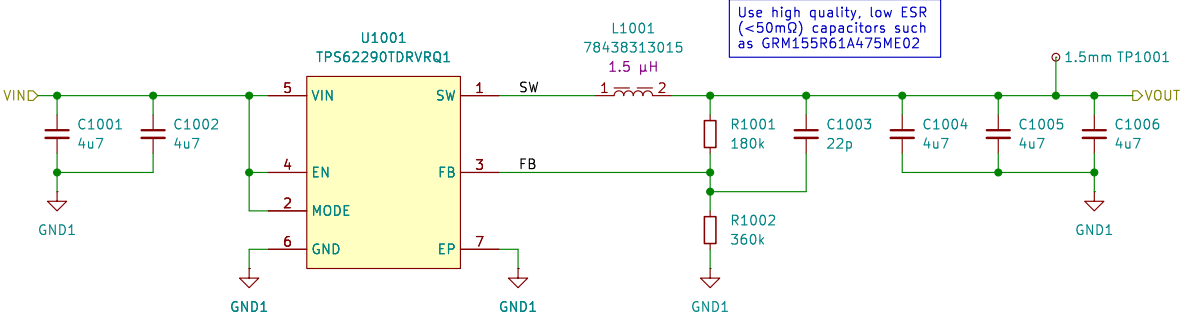
Date:

KiCad E.D.A. 9.0.7

Rev:

Id: 9/38

0V9 Regulator



0.9V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/0V9_1 Buck Converter/
File: 0v9_buck_converter.kicad_sch

Title:

Size: A4

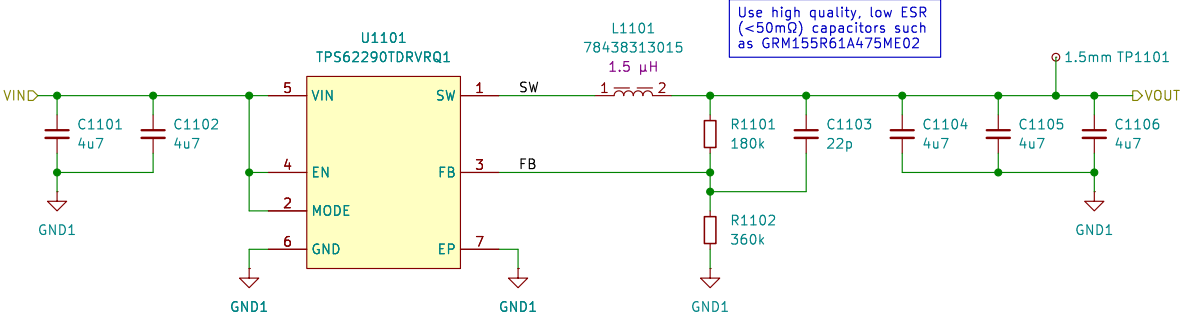
Date:

KiCad E.D.A. 9.0.7

Rev:

Id: 10/38

0V9 Regulator



0.9V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/0V9_2 Buck Converter/
File: 0v9_buck_converter.kicad_sch

Title:

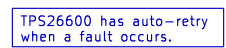
Size: A4

Date:

KiCad E.D.A. 9.0.7

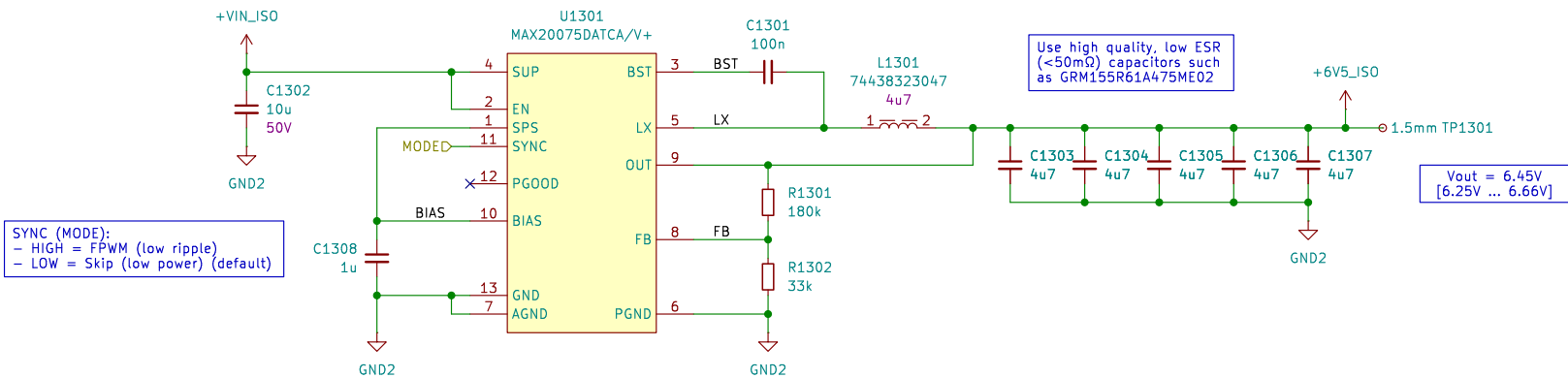
Rev:

Id: 11/38

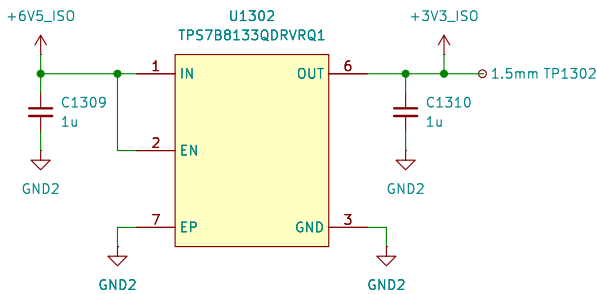


Id: 12/38

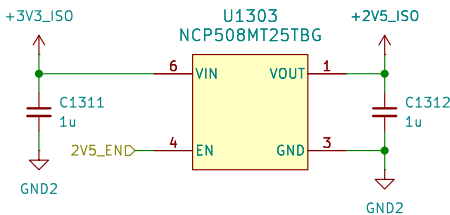
6V5 Regulator



3V3 Regulator



2V5 Regulator



6V5, 3V3 and 2V5 regulators for the connector side of the PCB.

Sheet: /Power/Regulators (Isolated)/
File: power-regulators-isolated.kicad_sch

Title: Regulators (Connector Side)

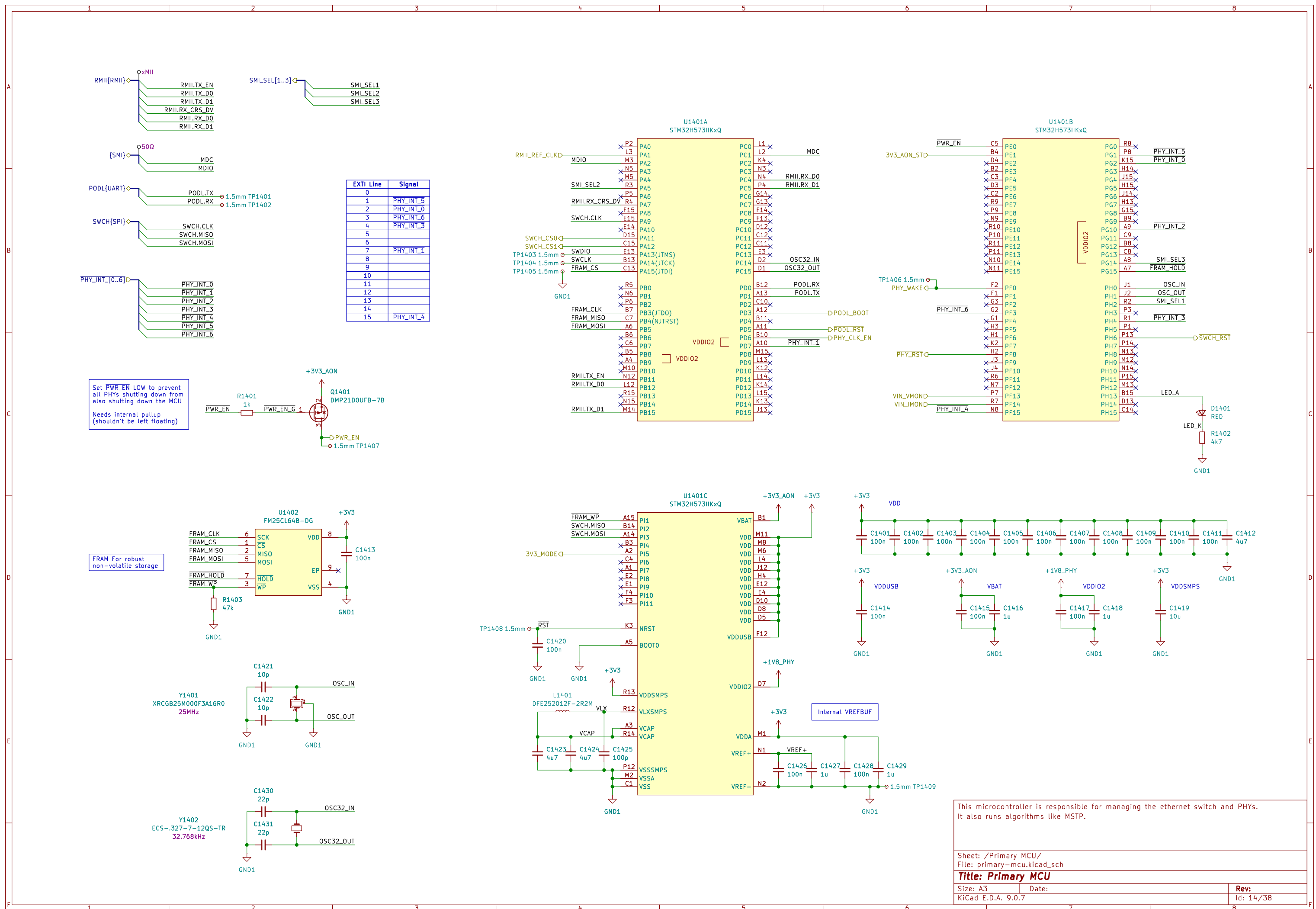
Size: A4

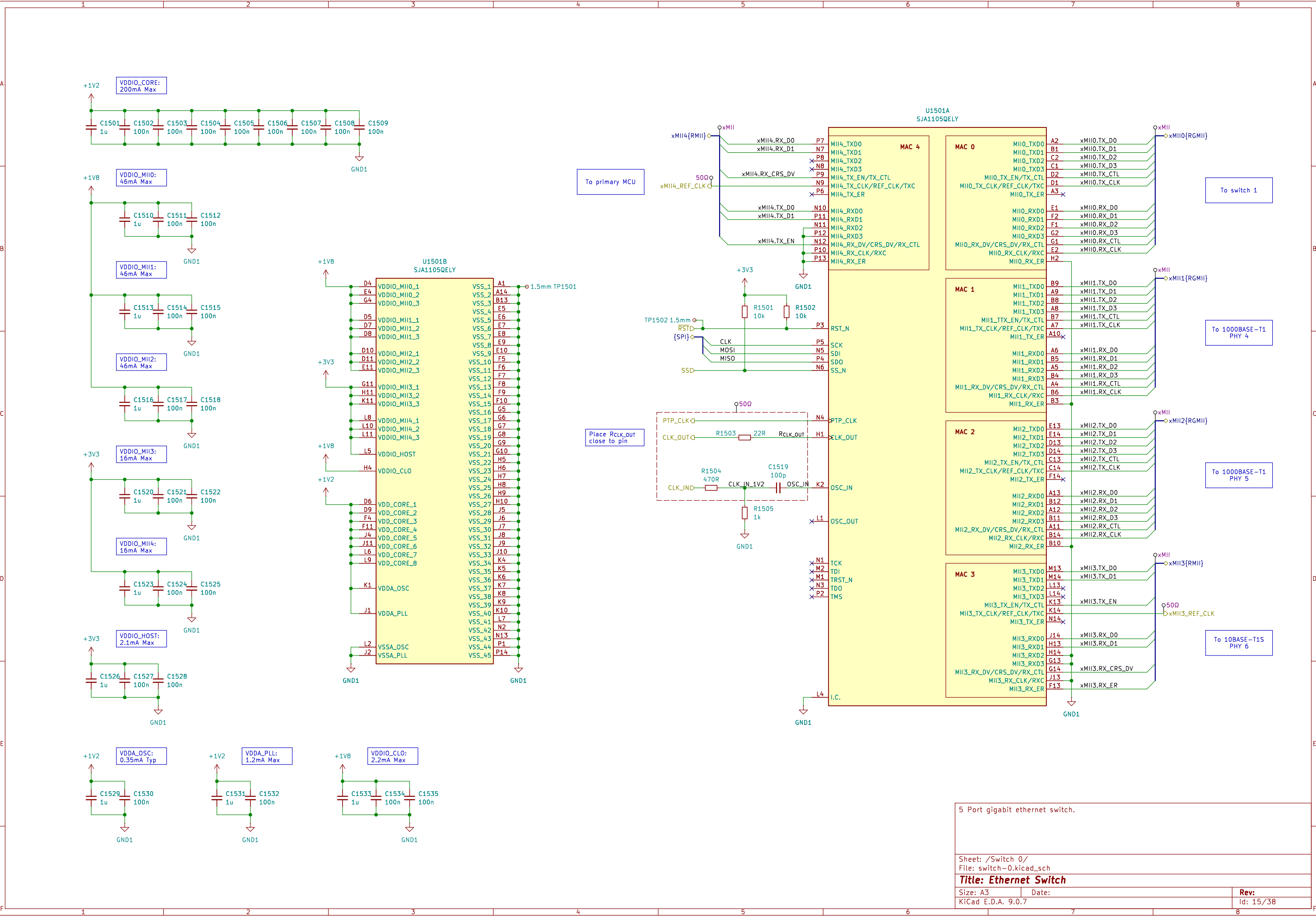
Date:

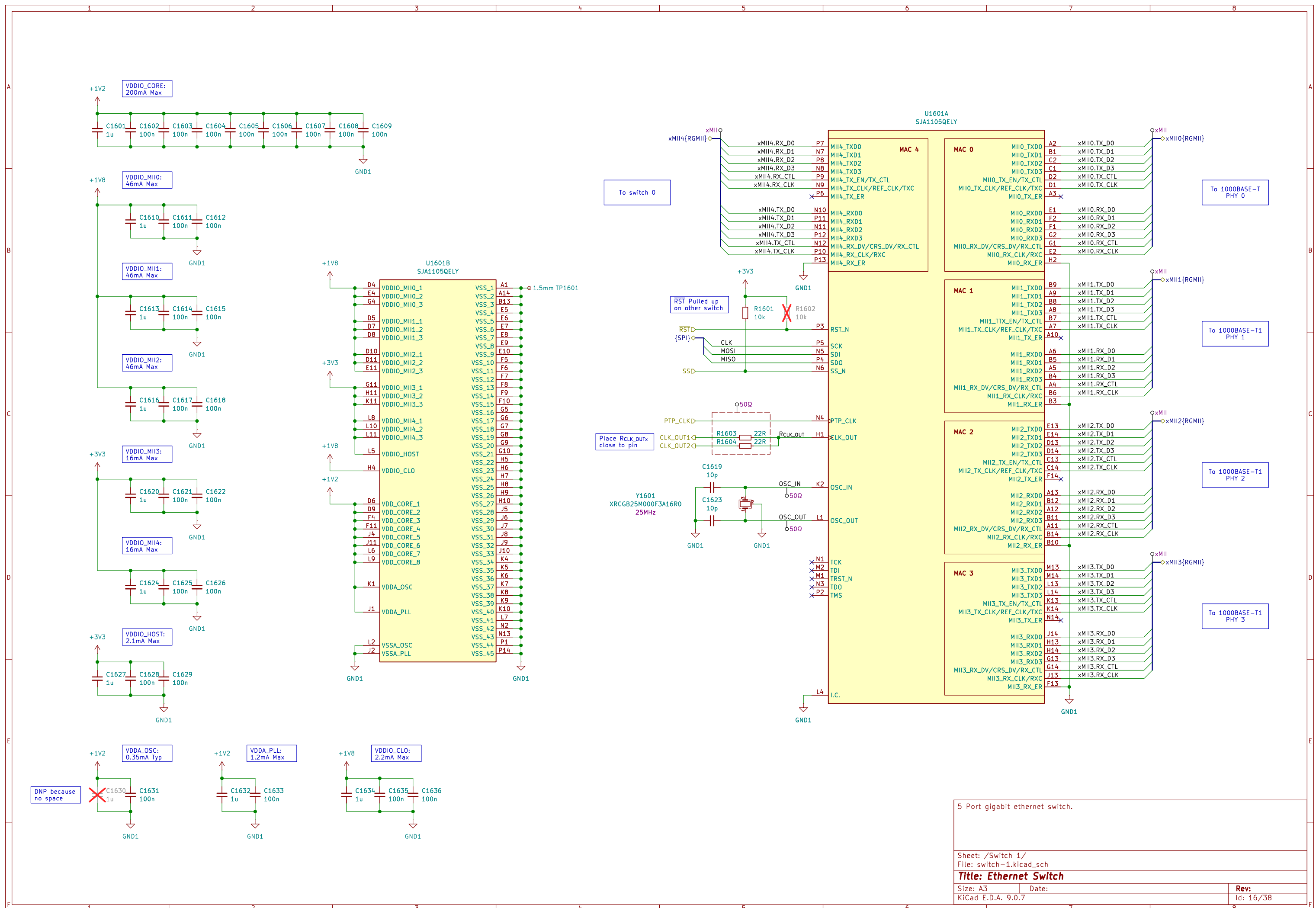
KiCad E.D.A. 9.0.7

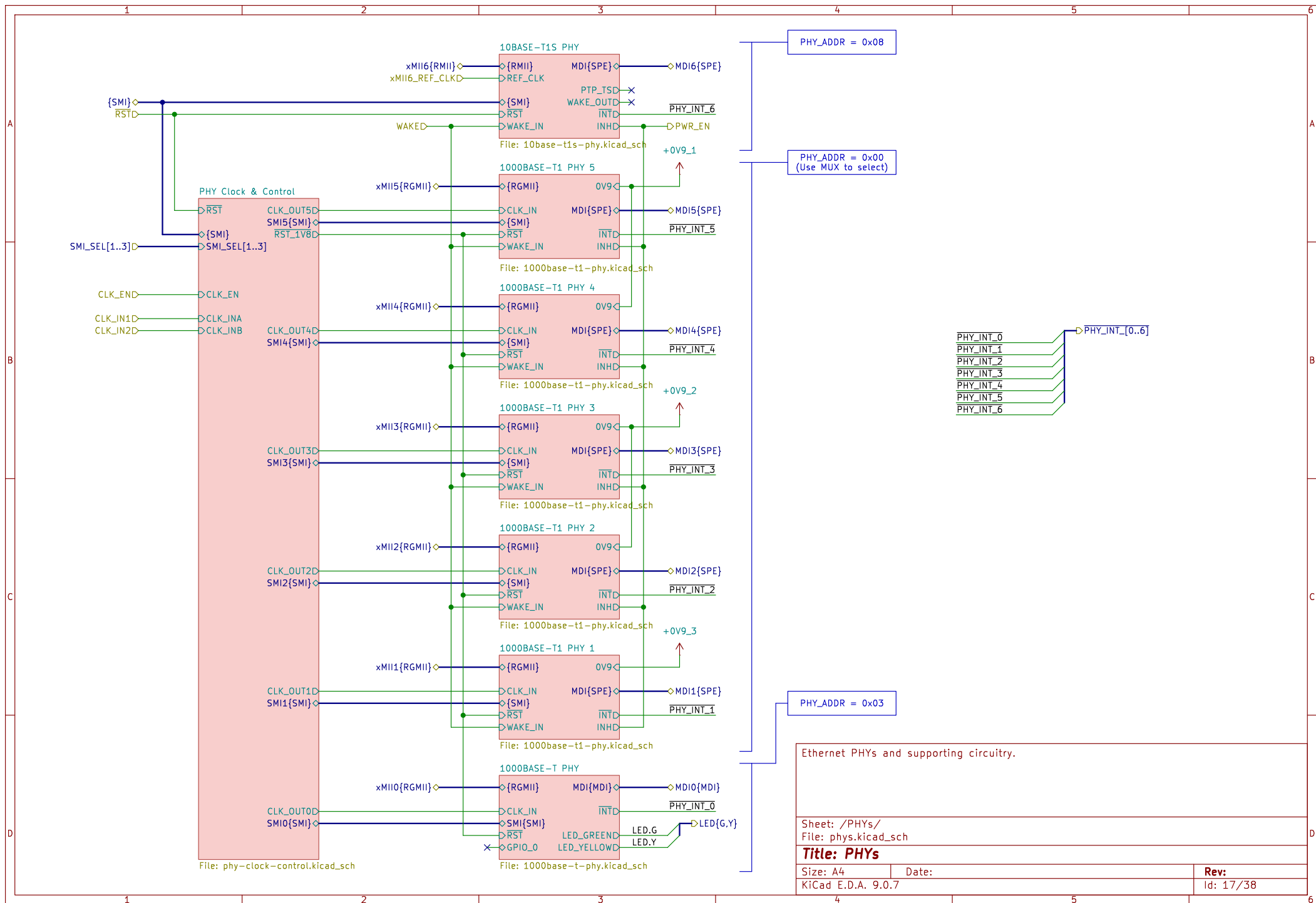
Rev:

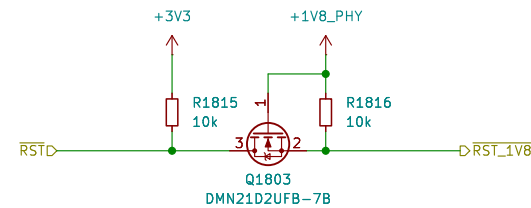
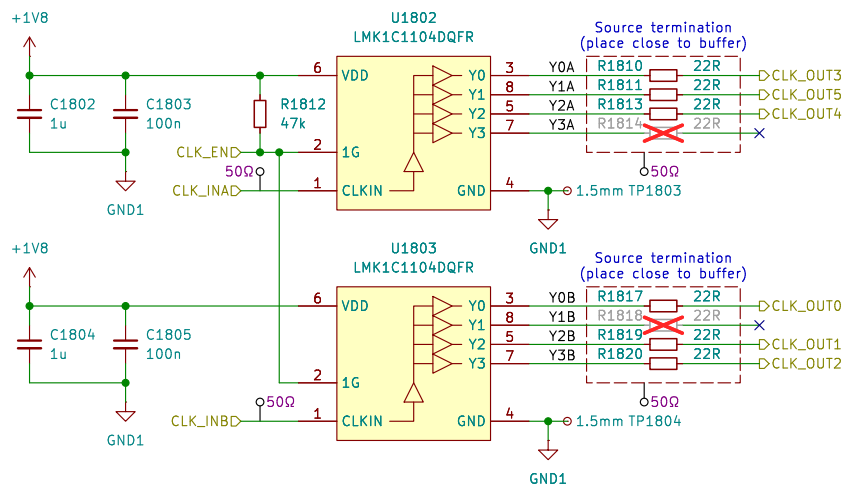
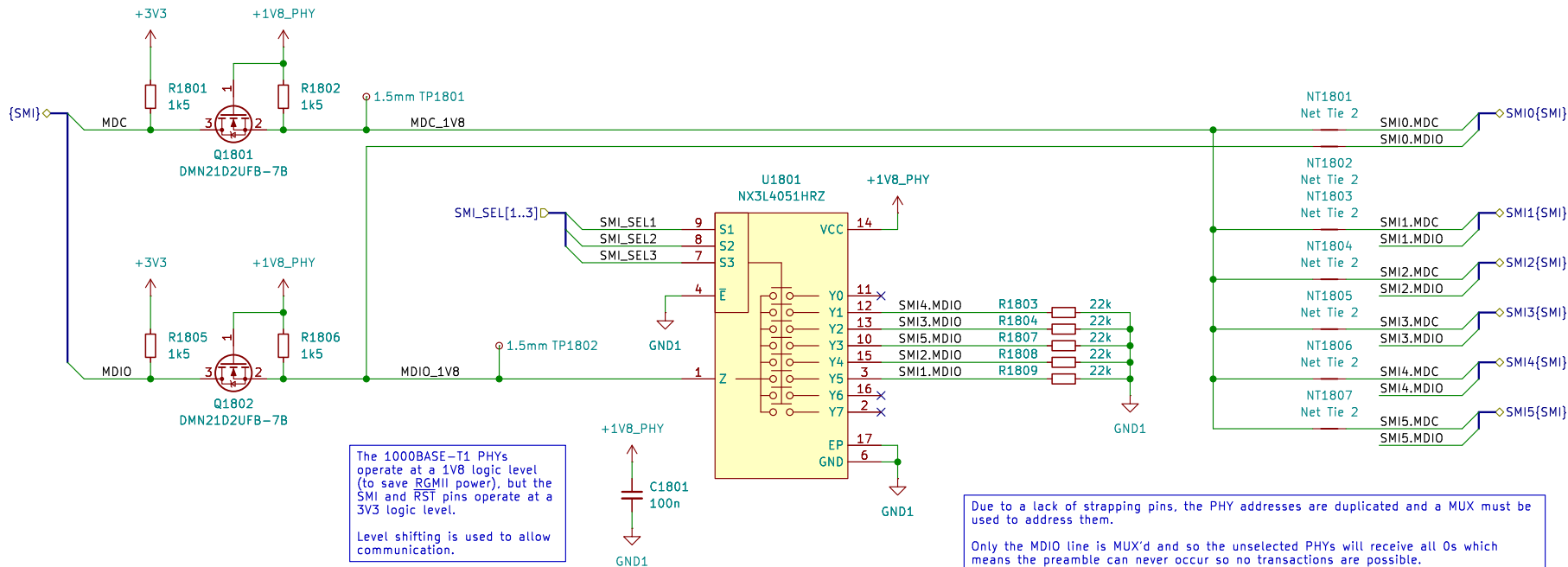
Id: 13/38











Converts 3.3V PHY management signals to 1.8V.

Sheet: /PHYs/PHY Clock & Control/
File: phy-clock-control.kicad_sch

Title: SMI Level Shifting

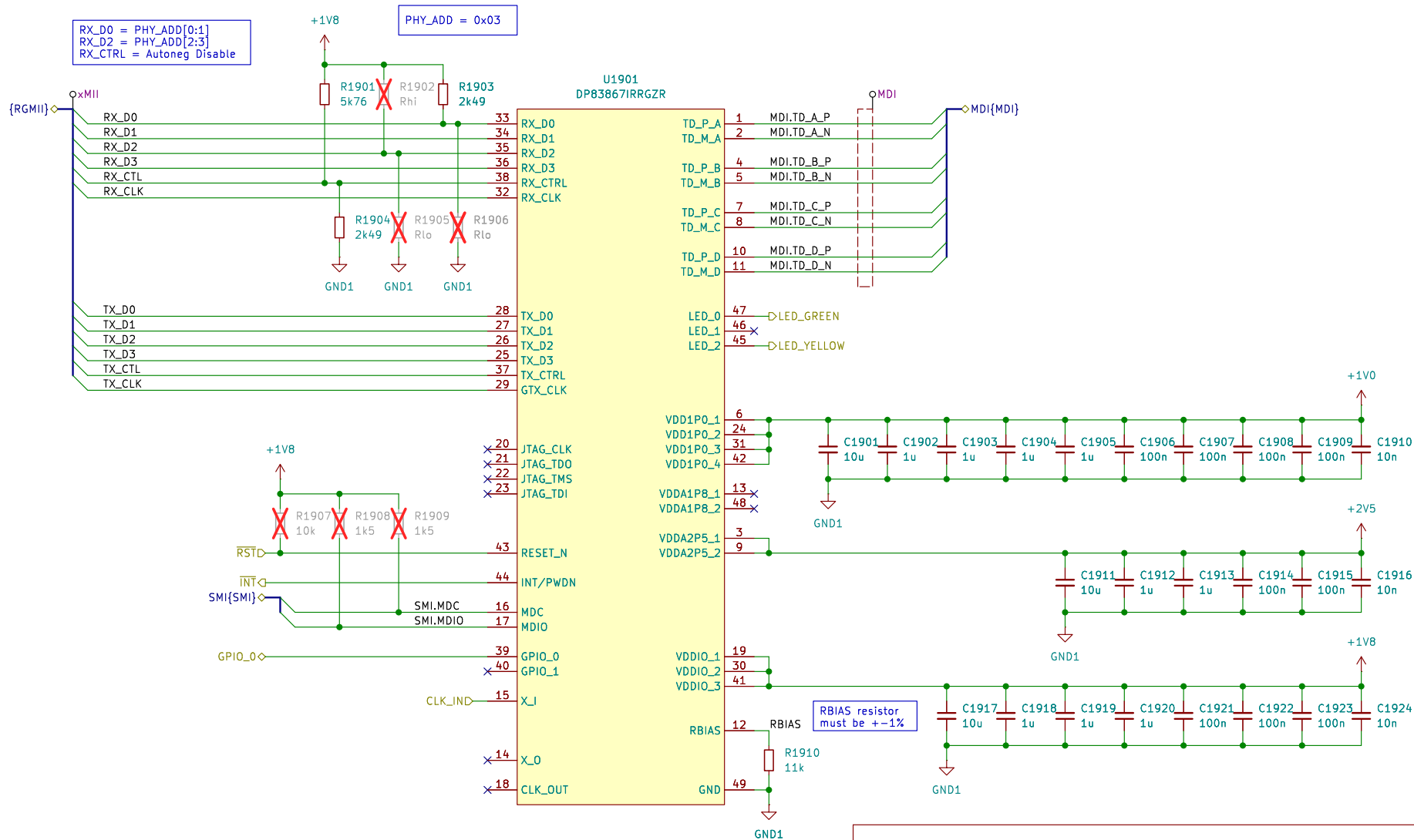
Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.7

Id: 18/38



Sheet: /PHYs/1000BASE-T PHY/
File: 1000base-t-phy.kicad_sch

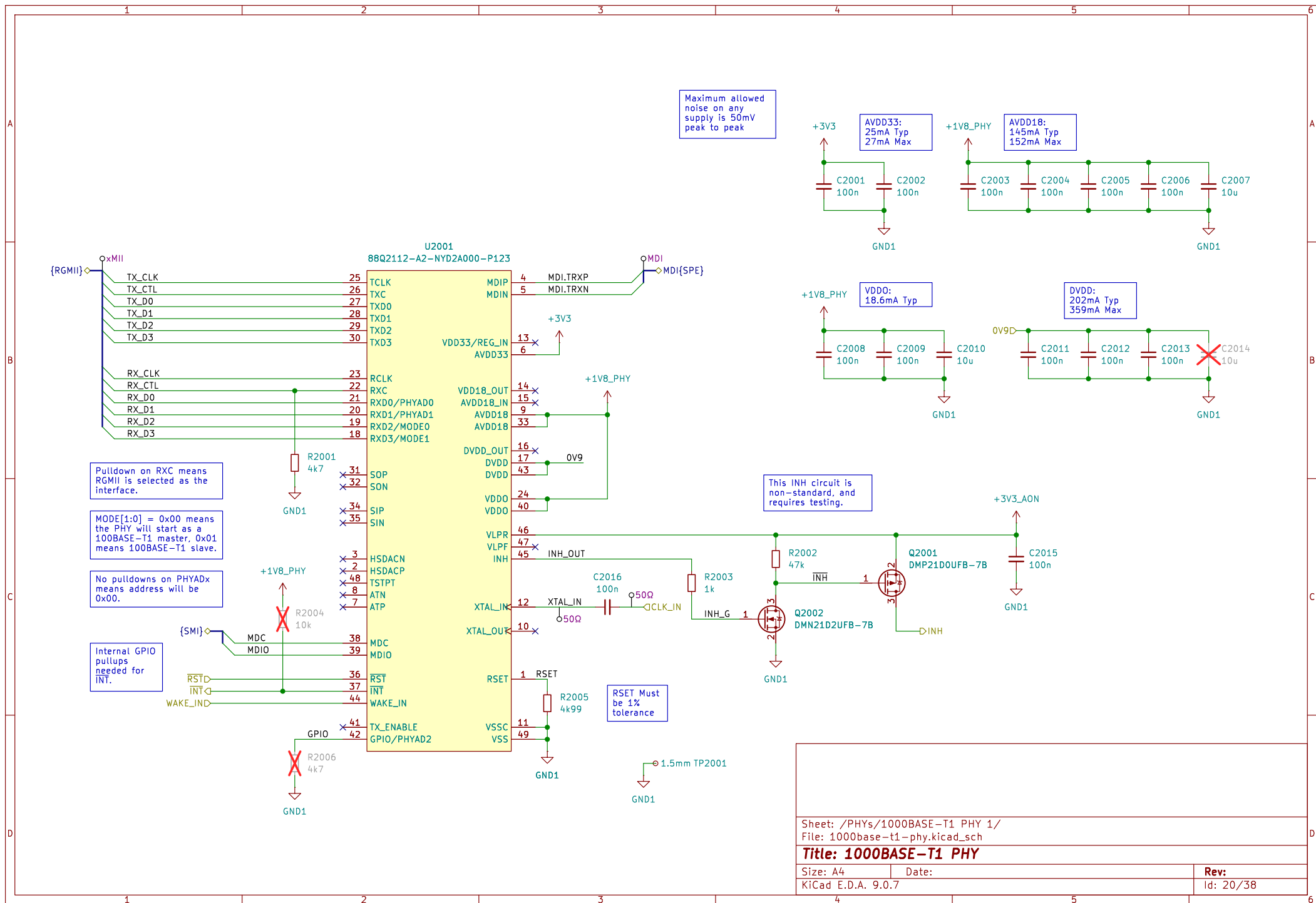
Title: PHY

Size: A4
KiCad E.D.A. 9.0.7

Date:

Rev: 1.0.0

Id: 19/38



Sheet: /PHYs/1000BASE-T1 PHY 1/
 File: 1000base-t1-phy.kicad_sch

Title: 1000BASE-T1 PHY

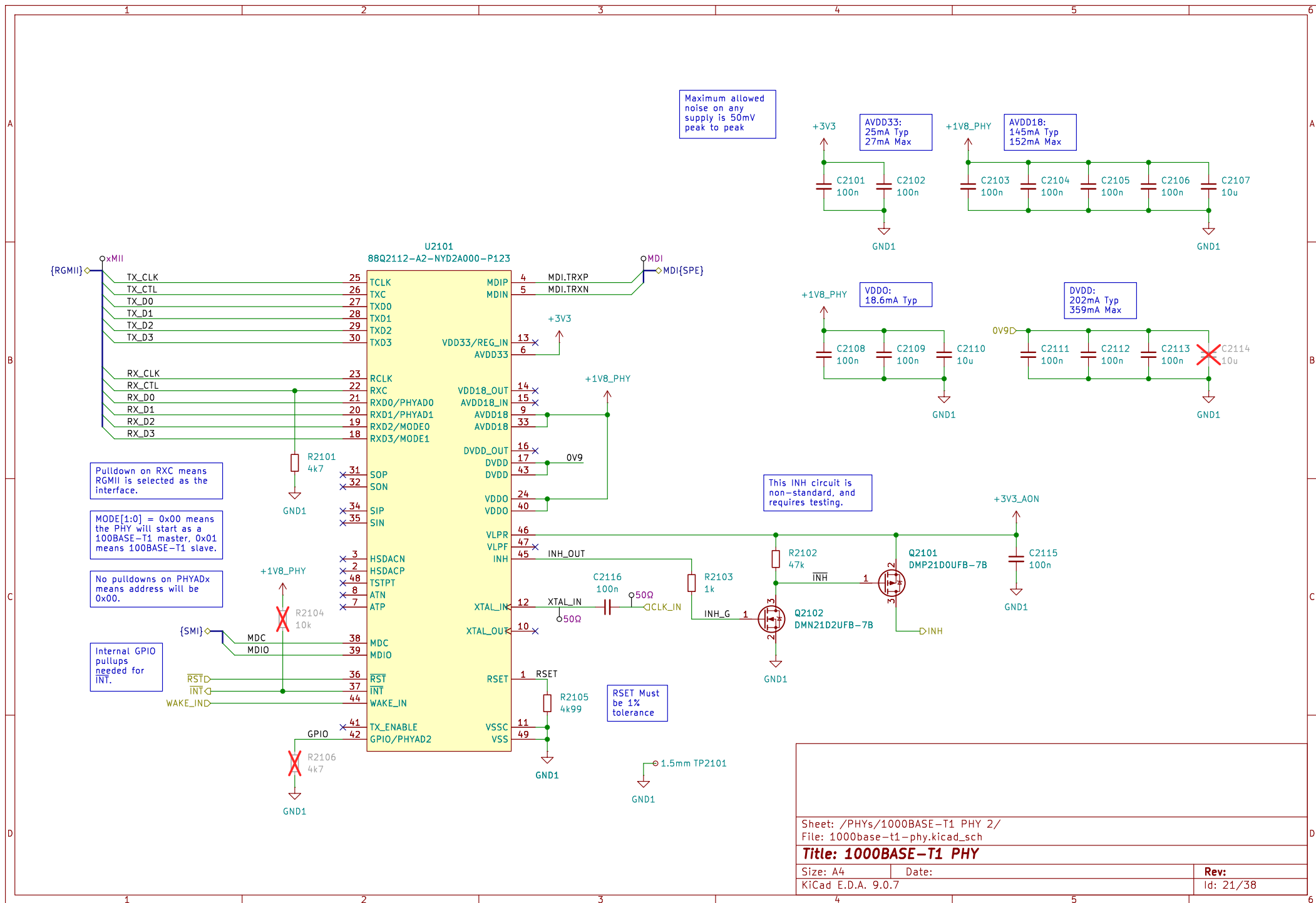
Size: A4

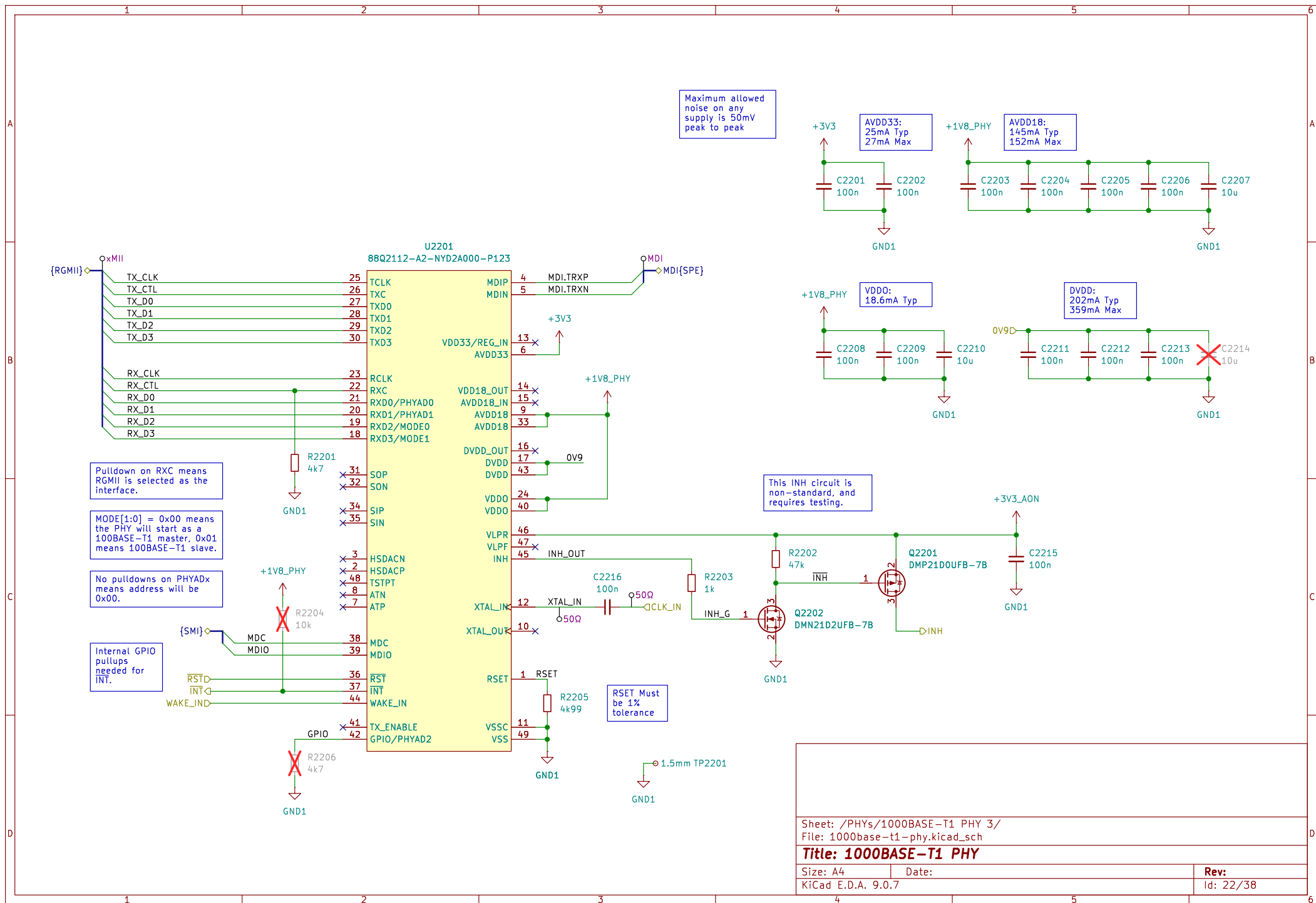
Date:

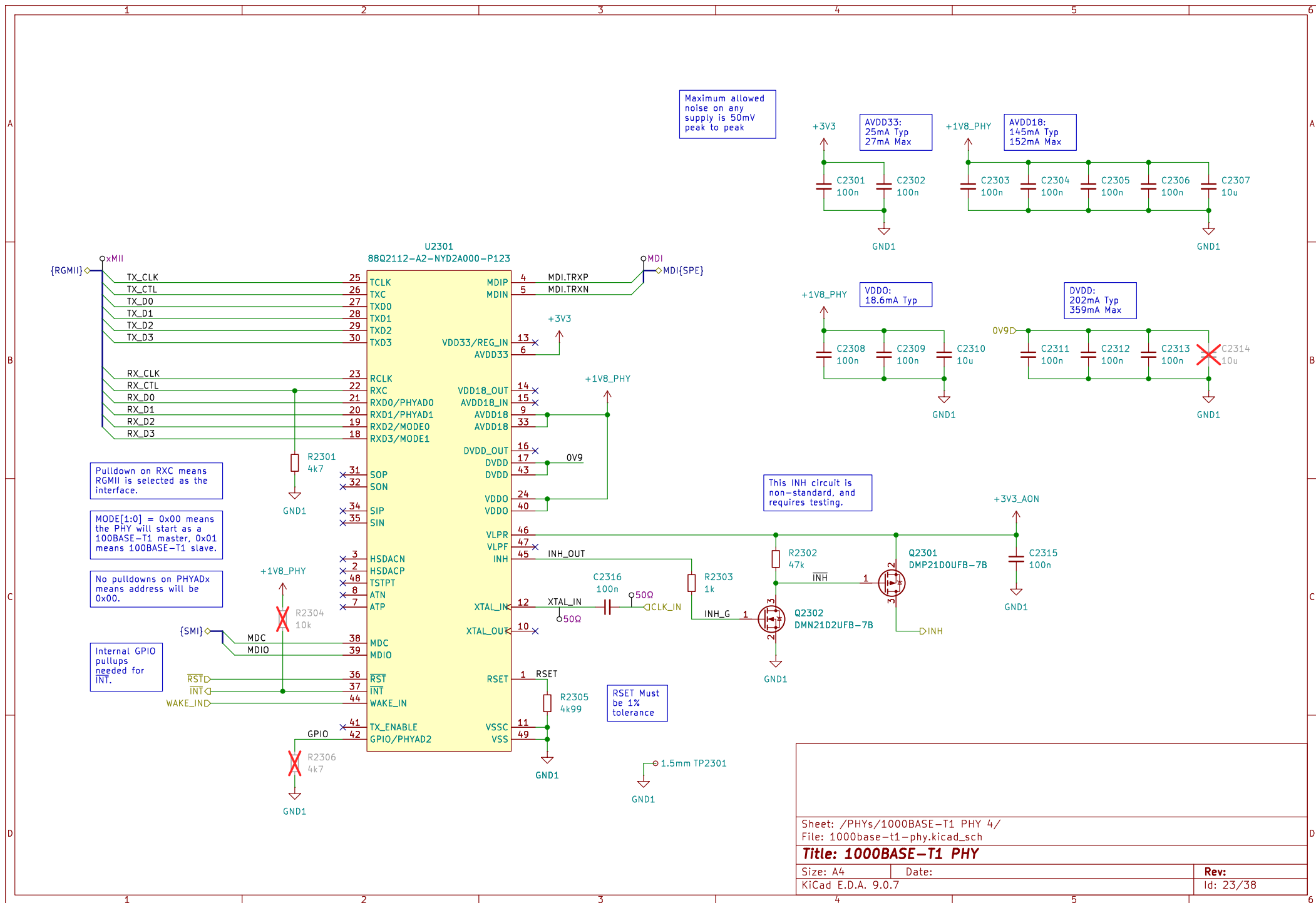
KiCad E.D.A. 9.0.7

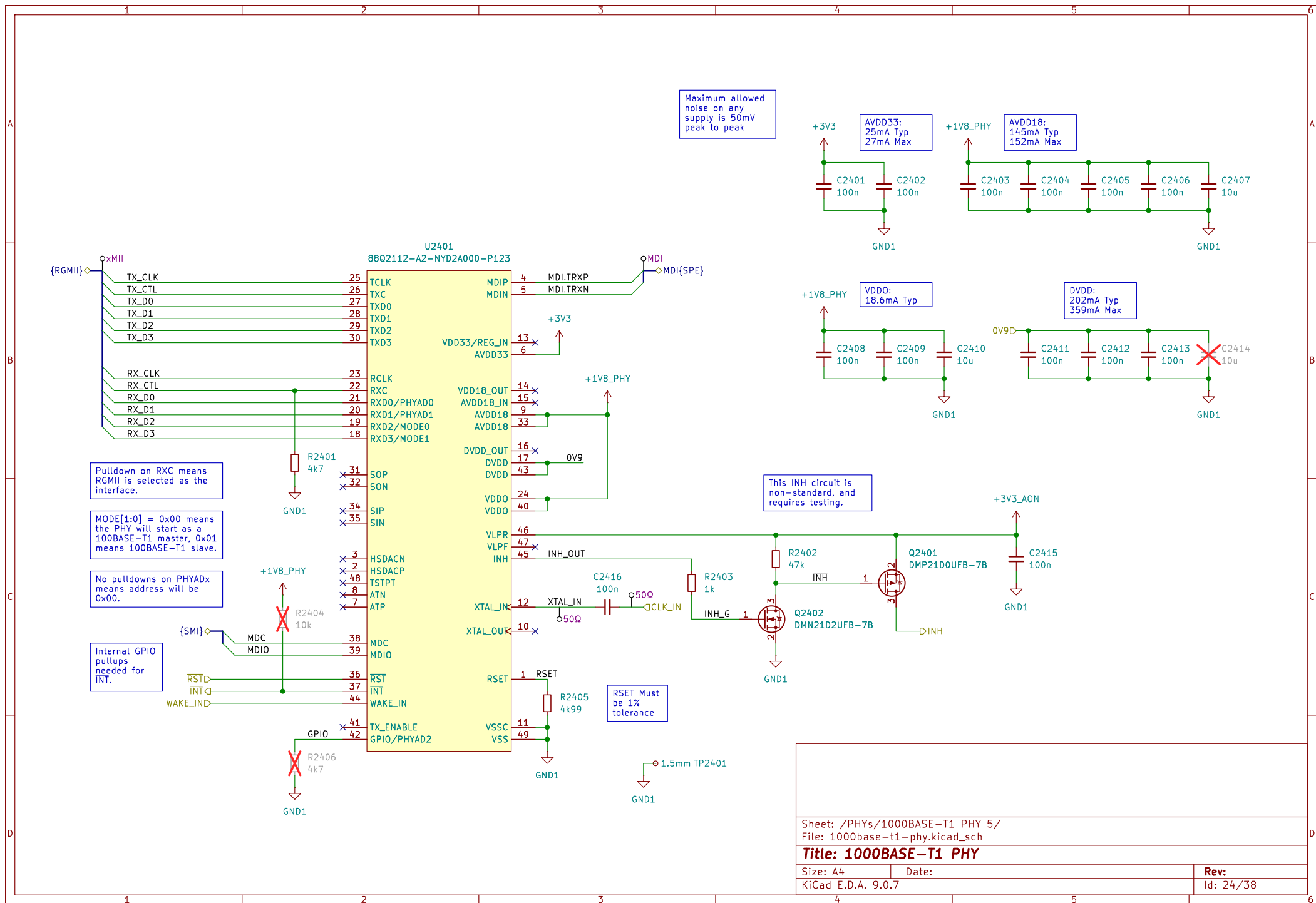
Rev:

Id: 20/38









Sheet: /PHYs/1000BASE-T1 PHY 5/
File: 1000base-t1-phy.kicad_sch

Title: 1000BASE-T1 PHY

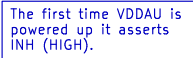
Size: A4

Date:

KiCad E.D.A. 9.0.7

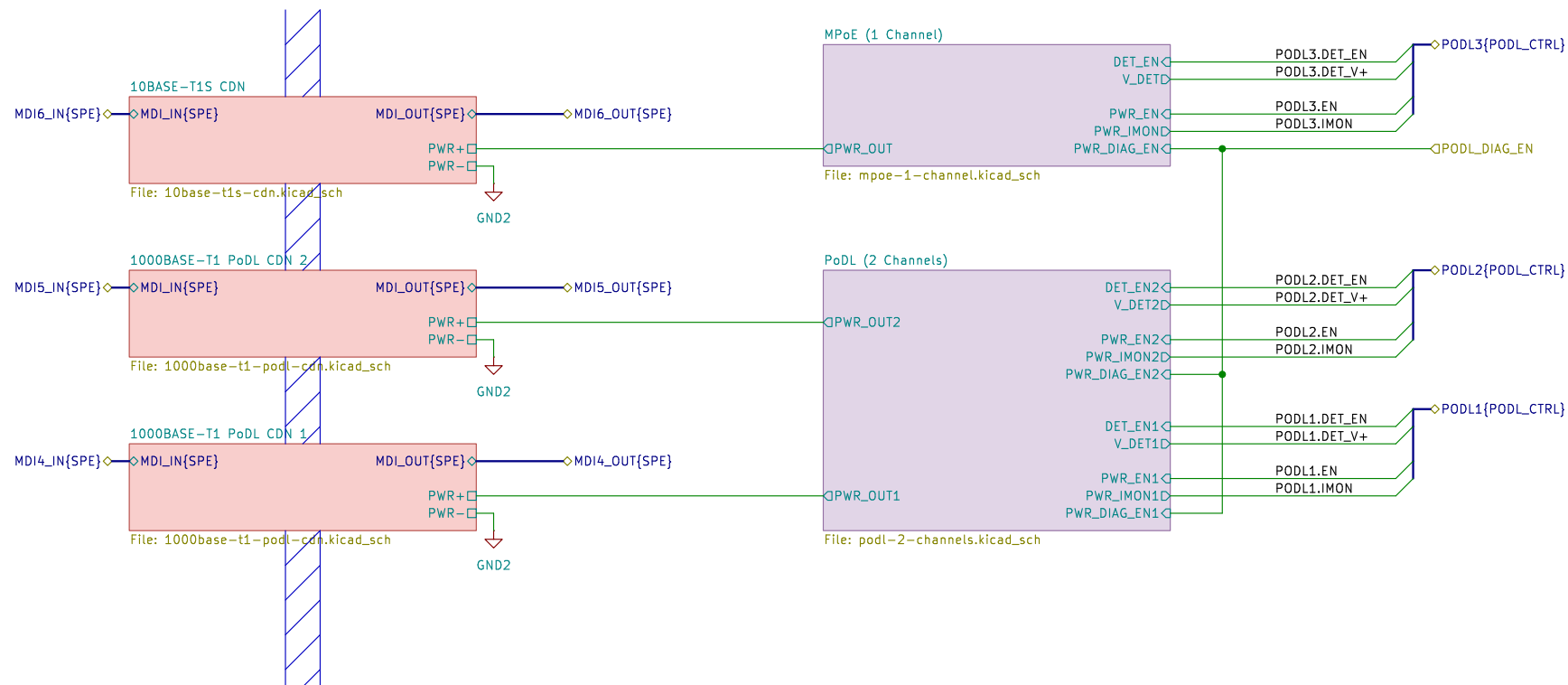
Rev:

Id: 24/38



Coupling-Decoupling Networks (CDNs)

Power Injection



Coupling-decoupling networks (CDNs) and power over datalines (PoDL).

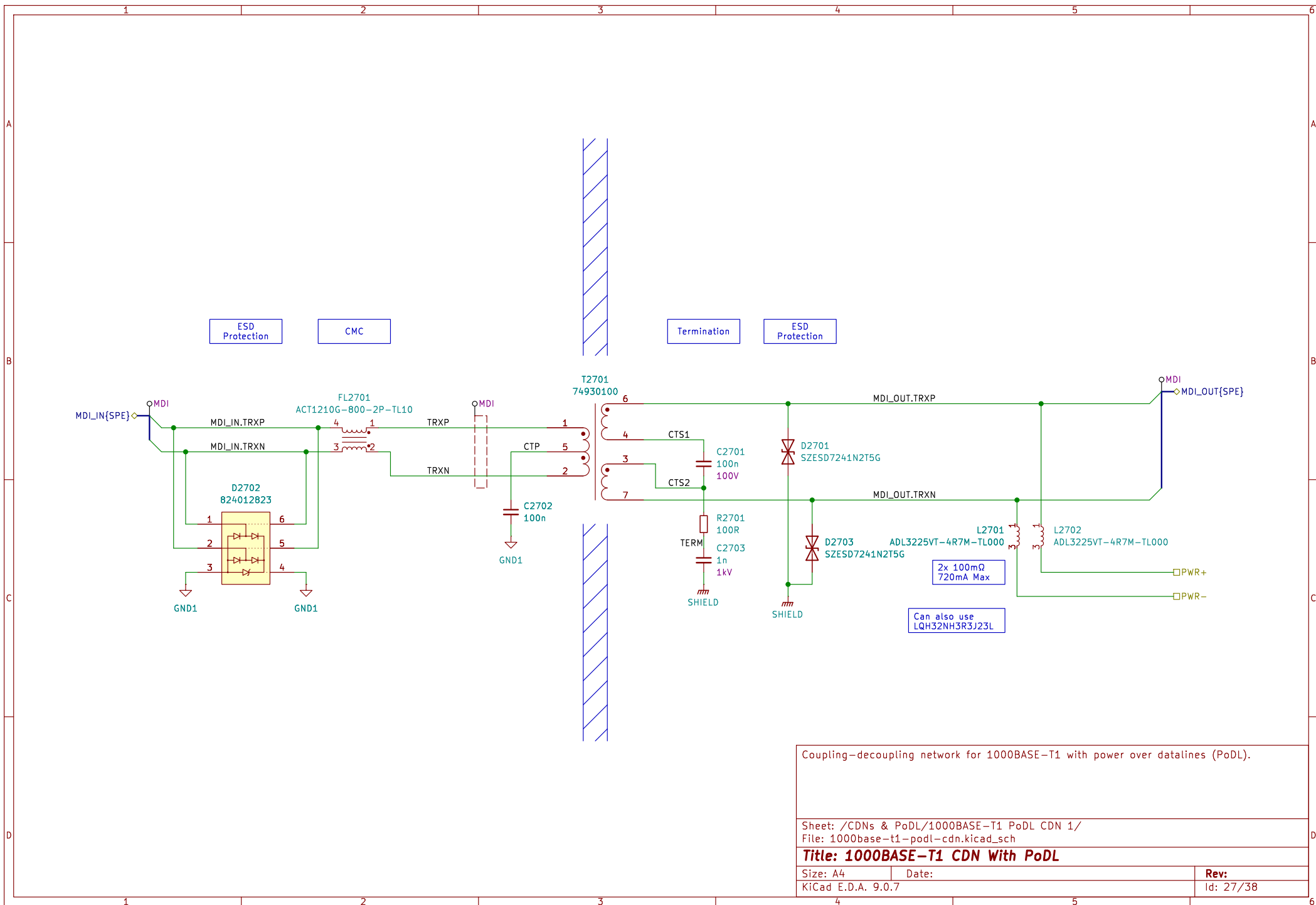
Sheet: /CDNs & PoDL/
File: cdn-podl.kicad_sch

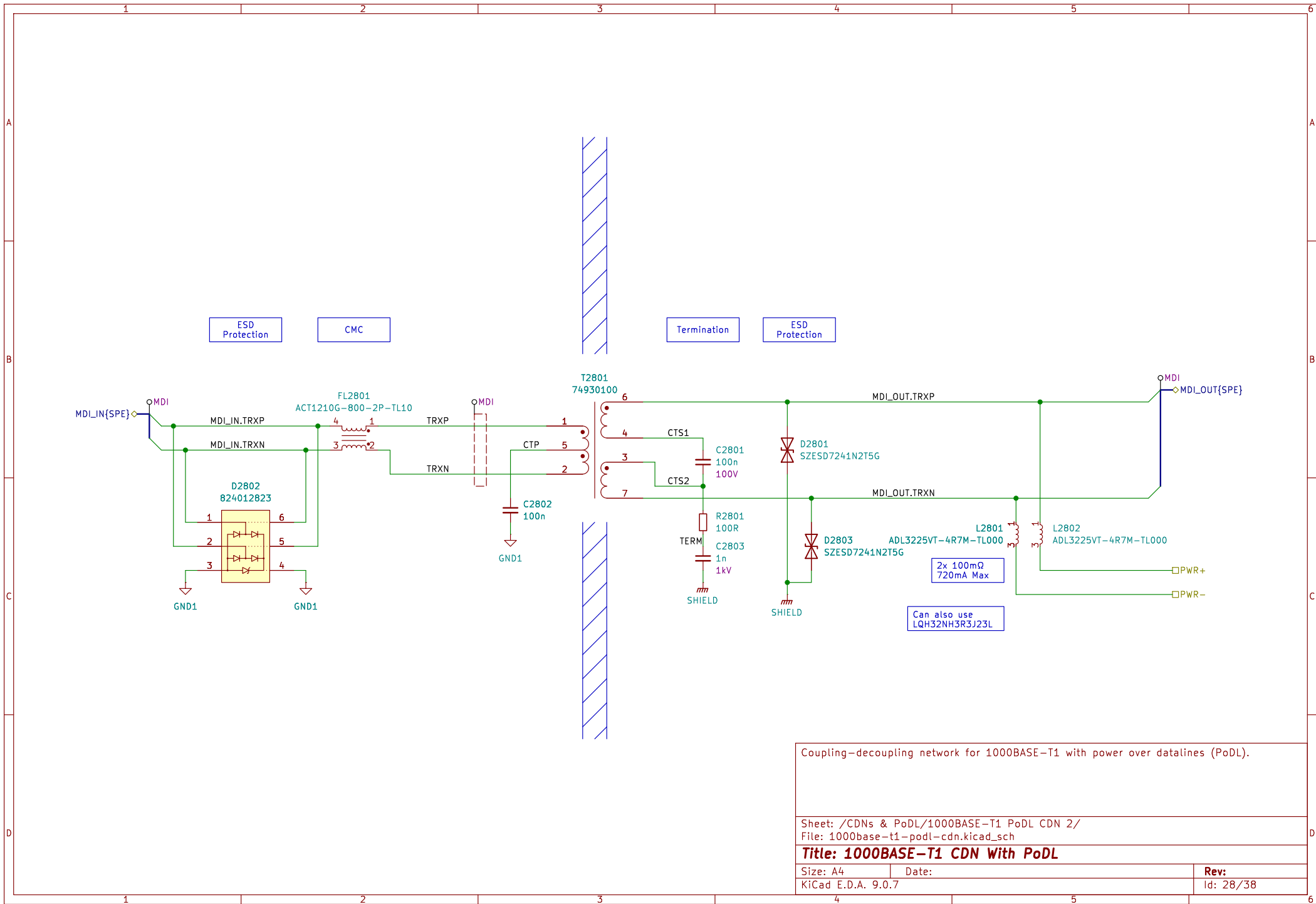
Title: CDNs and PoDL

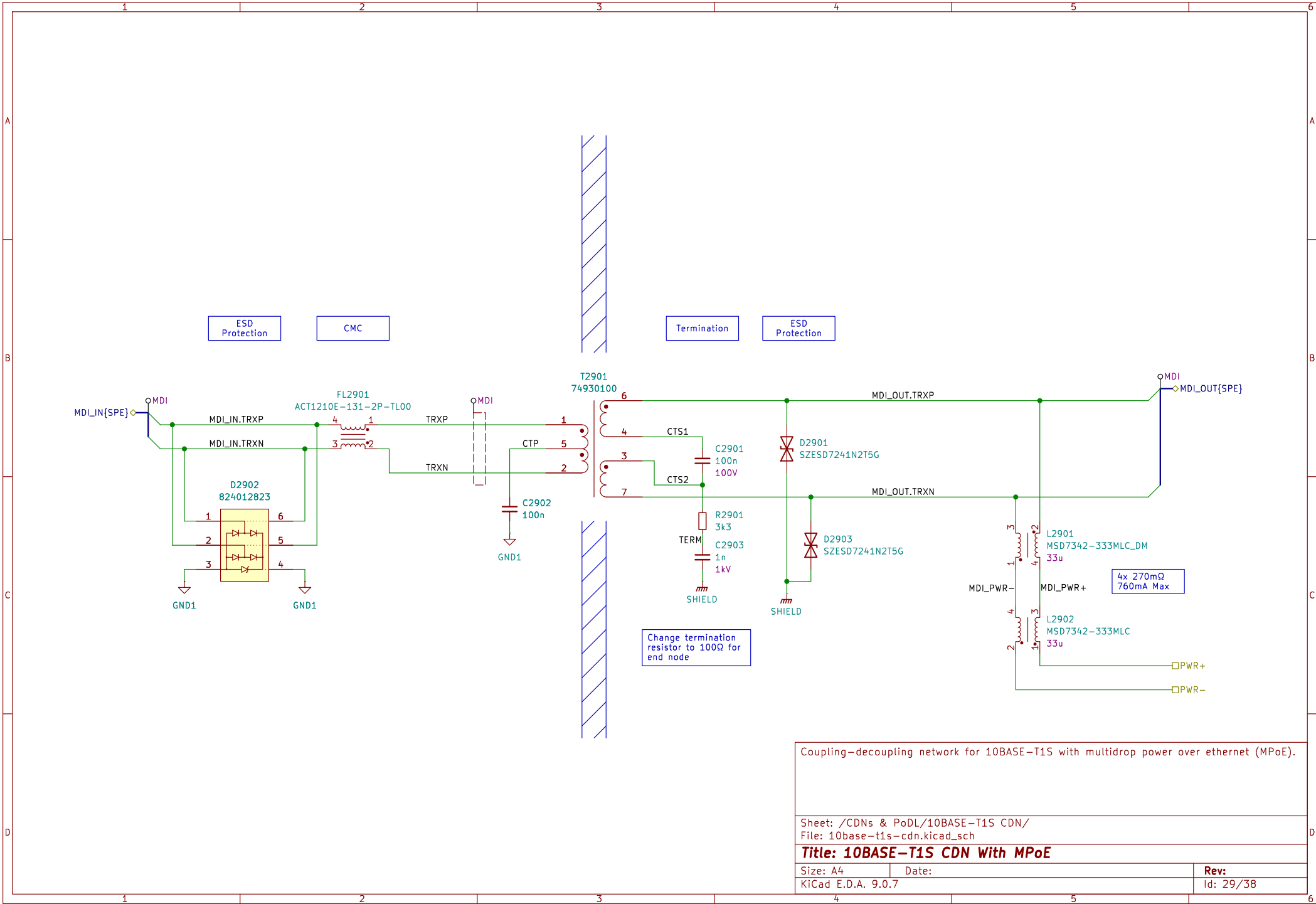
Size: A4
KiCad E.D.A. 9.0.7

Date:

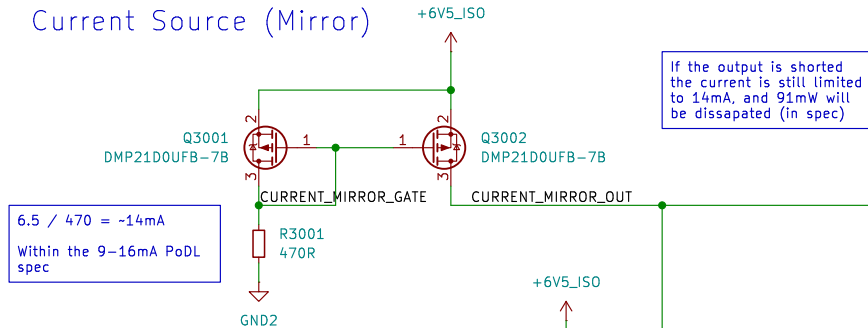
Rev:
Id: 26/38



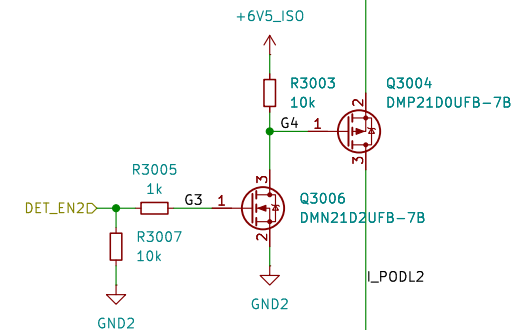
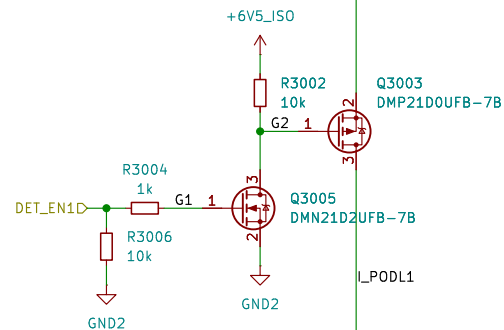




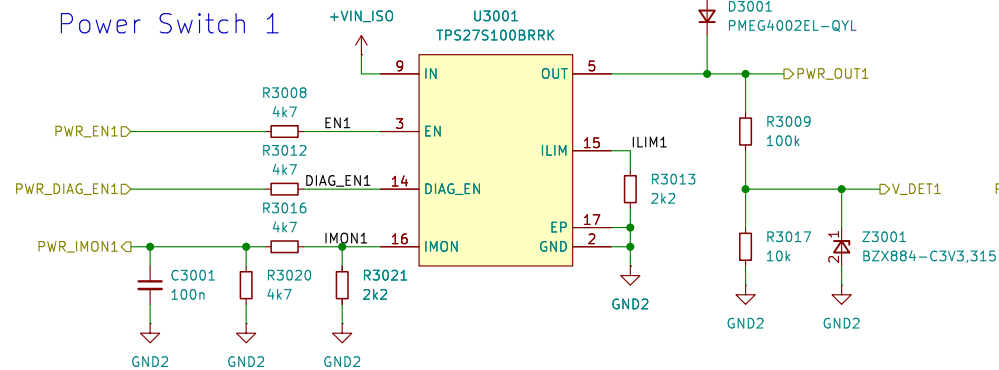
Current Source (Mirror)



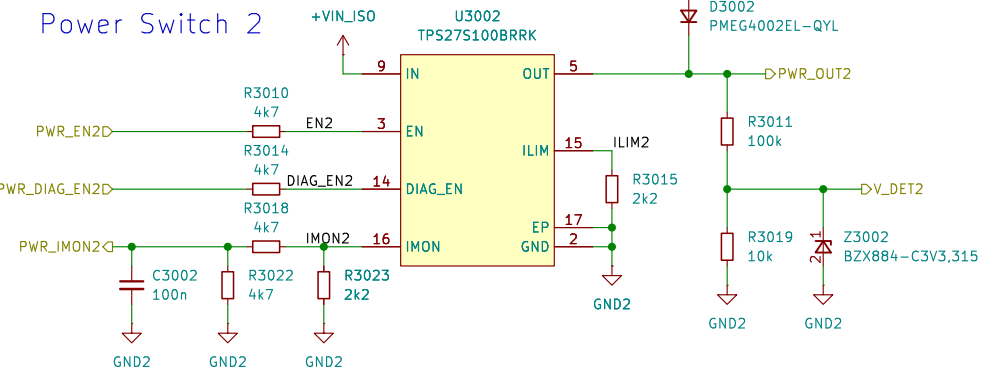
Current Switches



Power Switch 1



Power Switch 2



1.1A Maximum output current
1.1A Maximum measurable current

IMON can go up to 4.75V, potential divider limits this to be within 2.5V ADC range.

1/11 Potential divider and zener clamp to protect the 3V3 microcontroller IO.

$$I_{OUT} = 0.28046 \cdot V_{MON}$$

During the detection phase, 9-16mA is sent into TRXP and the resulting voltage is measured.
If a PoDL compatible device is detected on the other end (indicated by a 4.05-4.55V zener diode) then the full 24V is enabled.

Sheet: /CDNs & PoDL/PoDL (2 Channels)/
File: podl-2-channels.kicad_sch

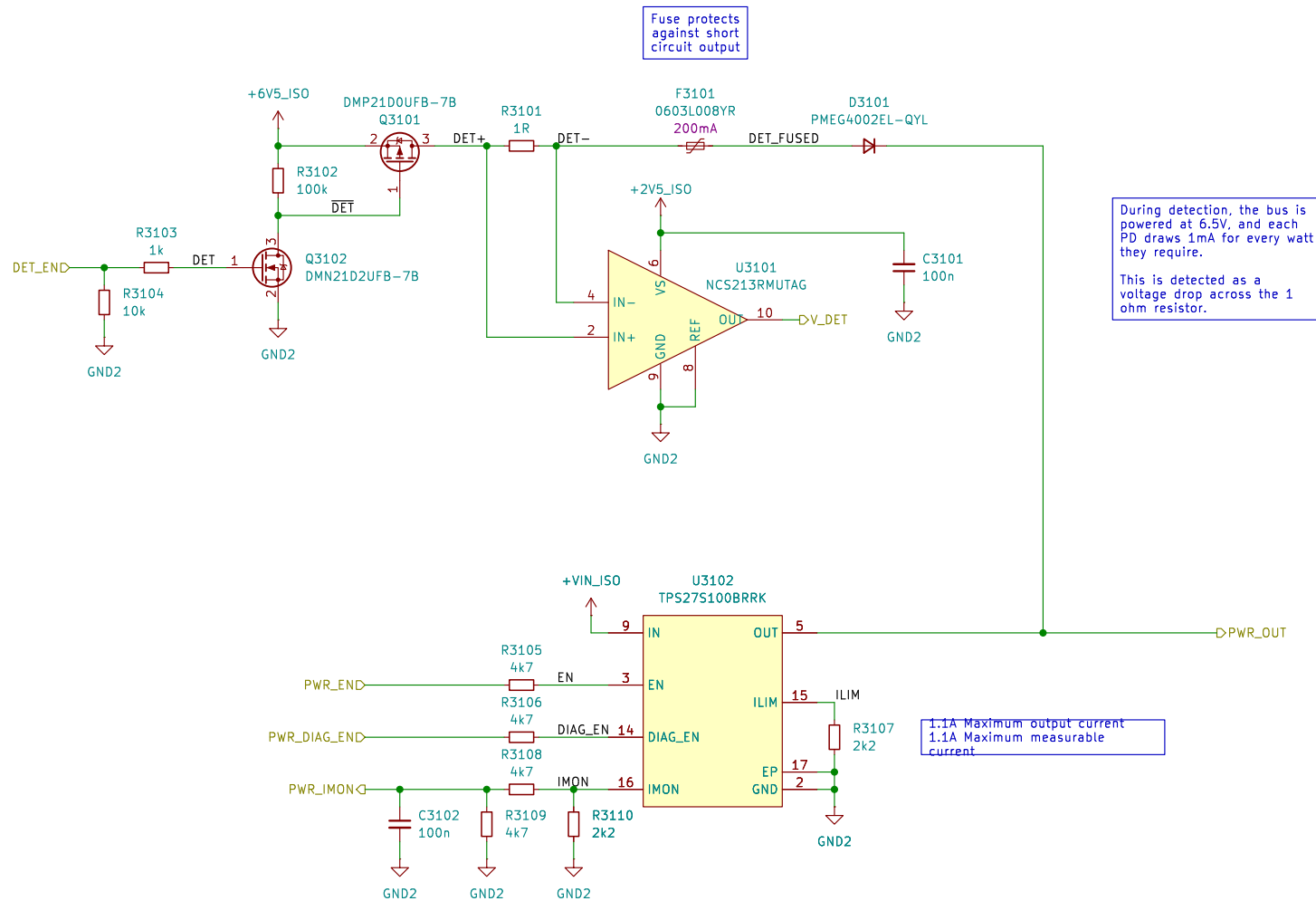
Title: PoDL Detection, Switching and Monitoring

Size: A4
KiCad E.D.A. 9.0.7

Date:

Rev:

Id: 30/38



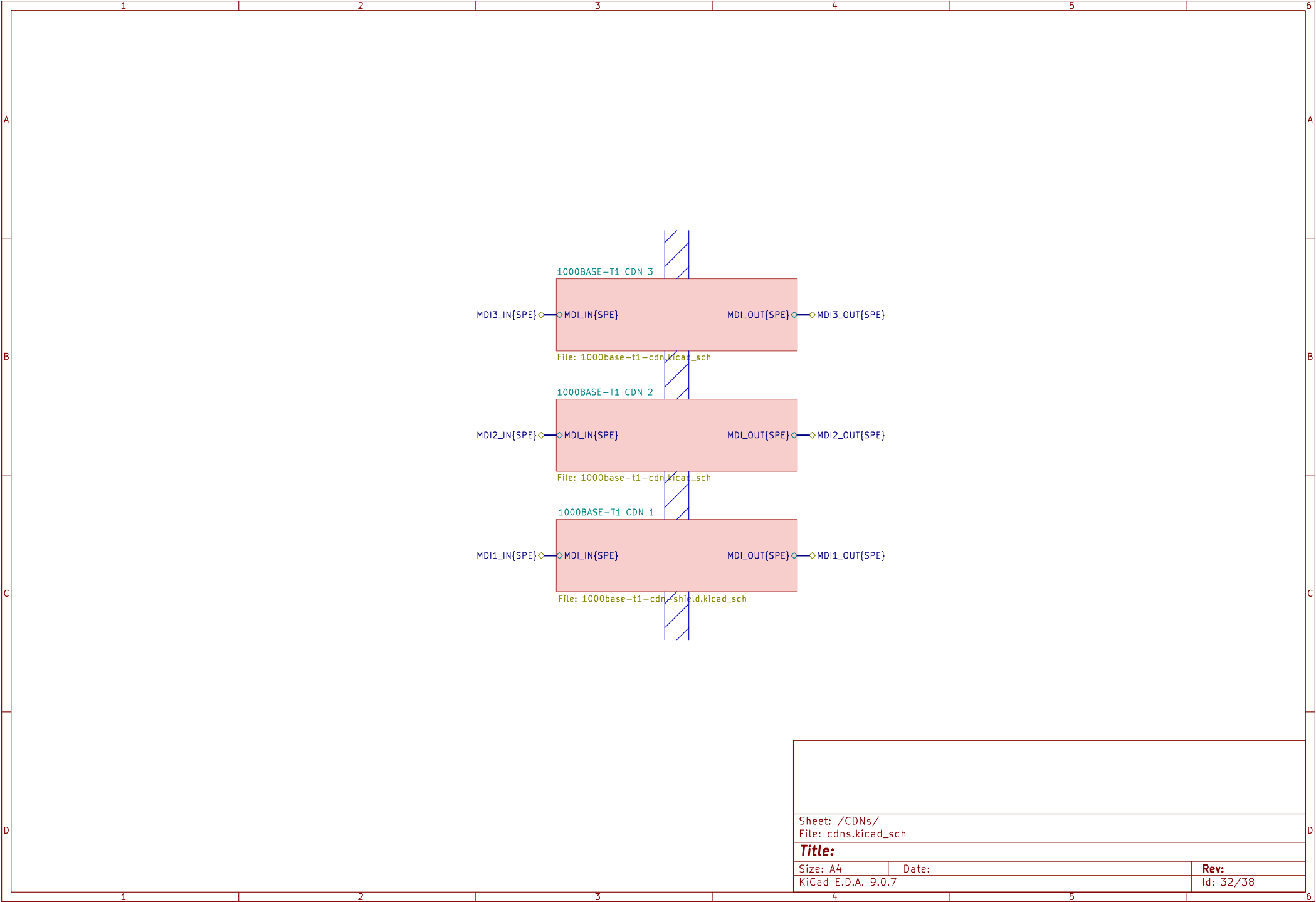
Sheet: /CDNs & PoDL/MPoE (1 Channel)/
File: mpoe-1-channel.kicad_sch

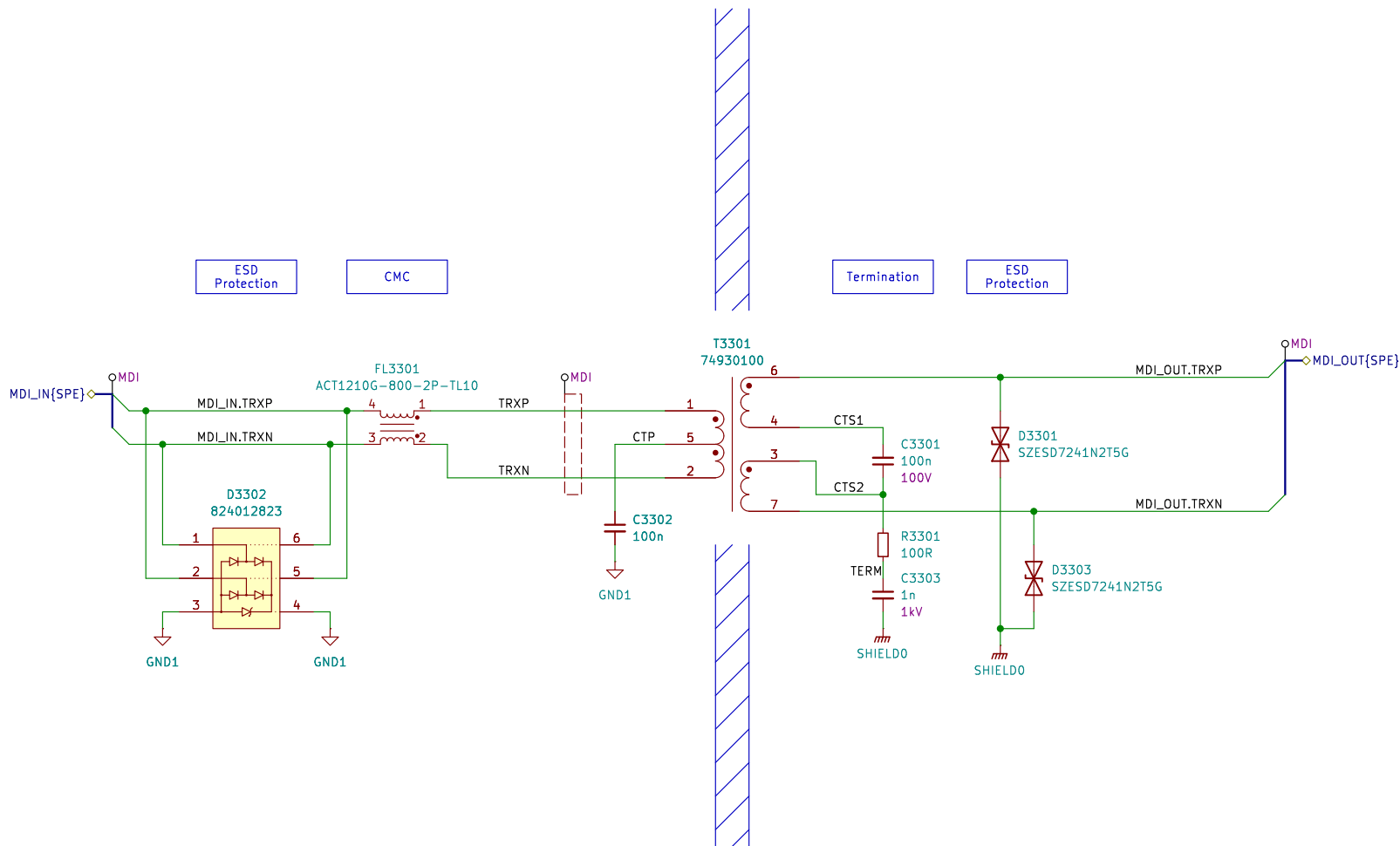
Title: MPoE Detection, Switching and Monitoring

Size: A4
KiCad E.D.A. 9.0.7

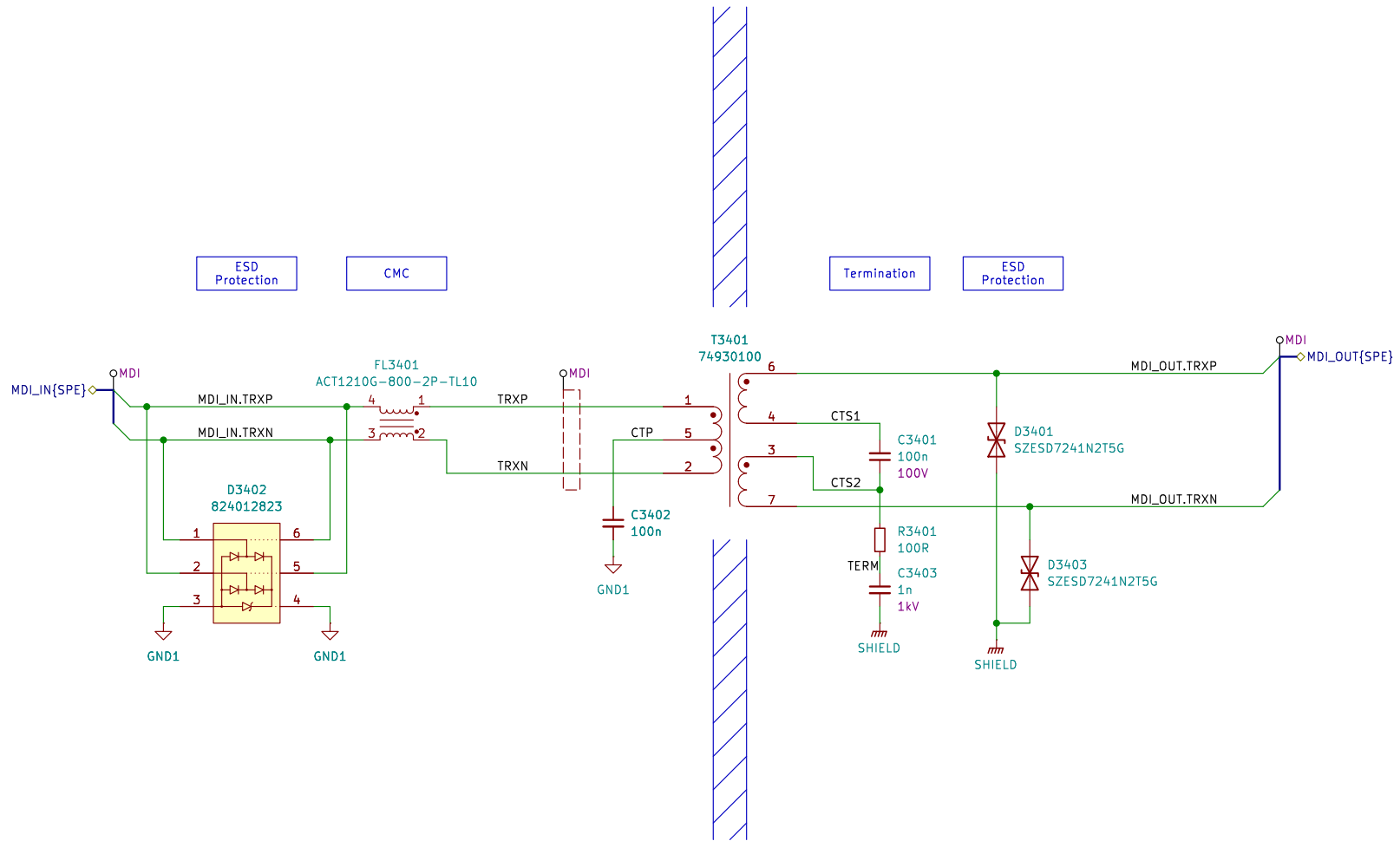
Date:

Rev:
Id: 31/38





Coupling-decoupling network for 1000BASE-T1		
Sheet: /CDNs/1000BASE-T1 CDN 1/		
File: 1000base-t1-cdn-shield.kicad_sch		
Title: 1000BASE-T1 CDN		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 33/38



Coupling-decoupling network for 1000BASE-T1

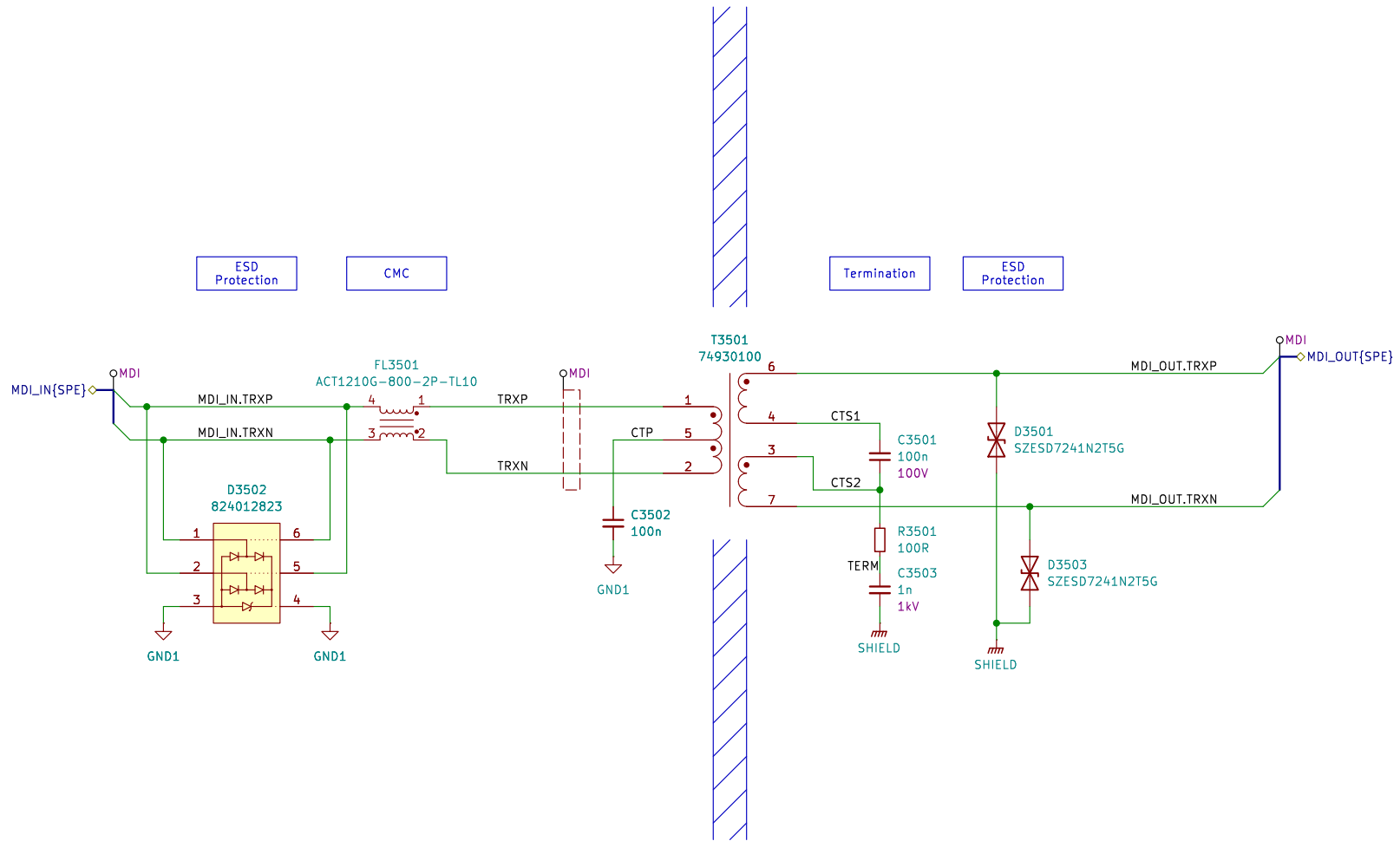
Sheet: /CDNs/1000BASE-T1 CDN 2/
File: 1000base-t1-cdn.kicad_sch

Title: 1000BASE-T1 CDN

Size: A4
KiCad E.D.A. 9.0.7

Date:

Rev:
Id: 34/38



Coupling-decoupling network for 1000BASE-T1

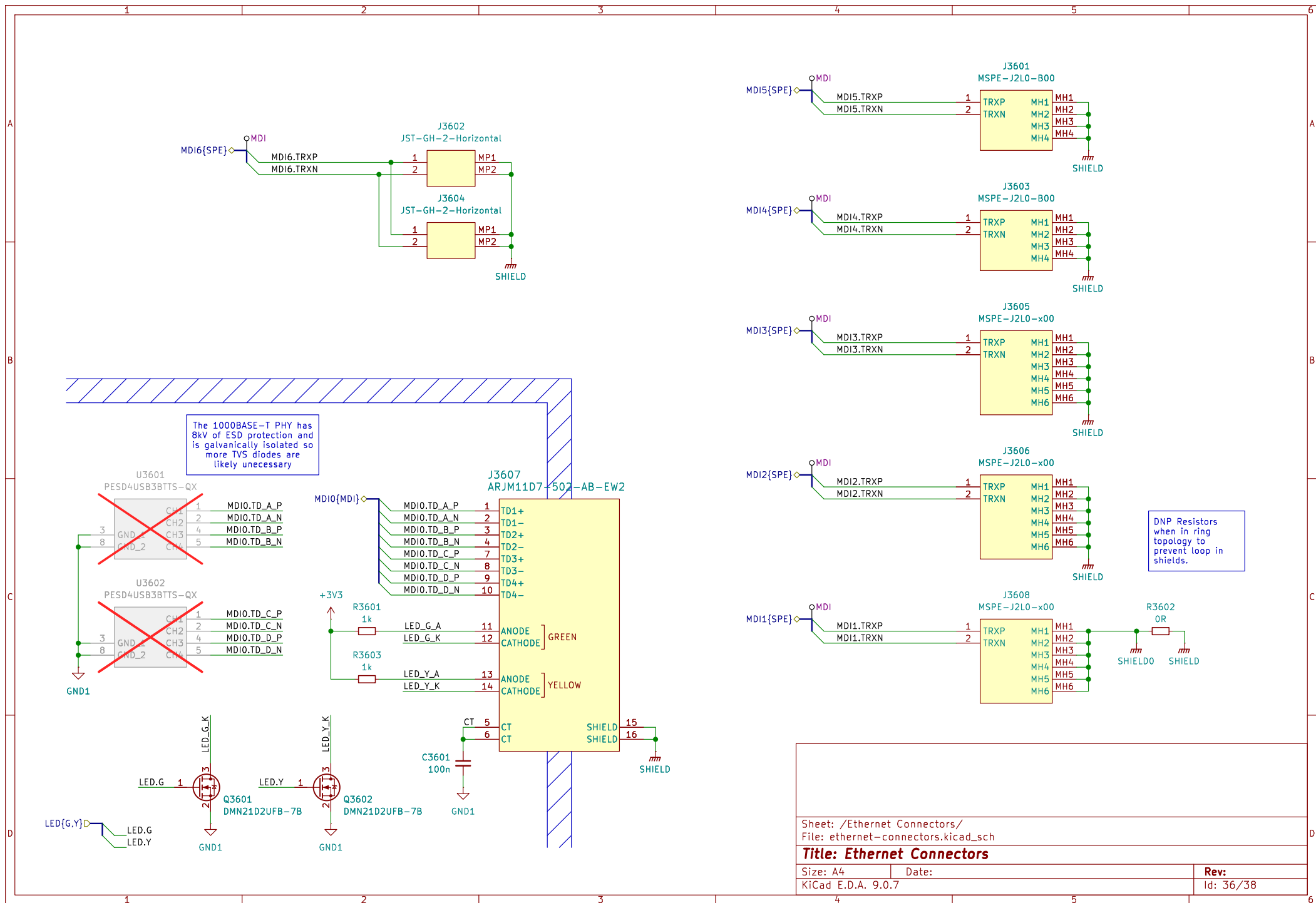
Sheet: /CDNs/1000BASE-T1 CDN 3/
File: 1000base-t1-cdn.kicad_sch

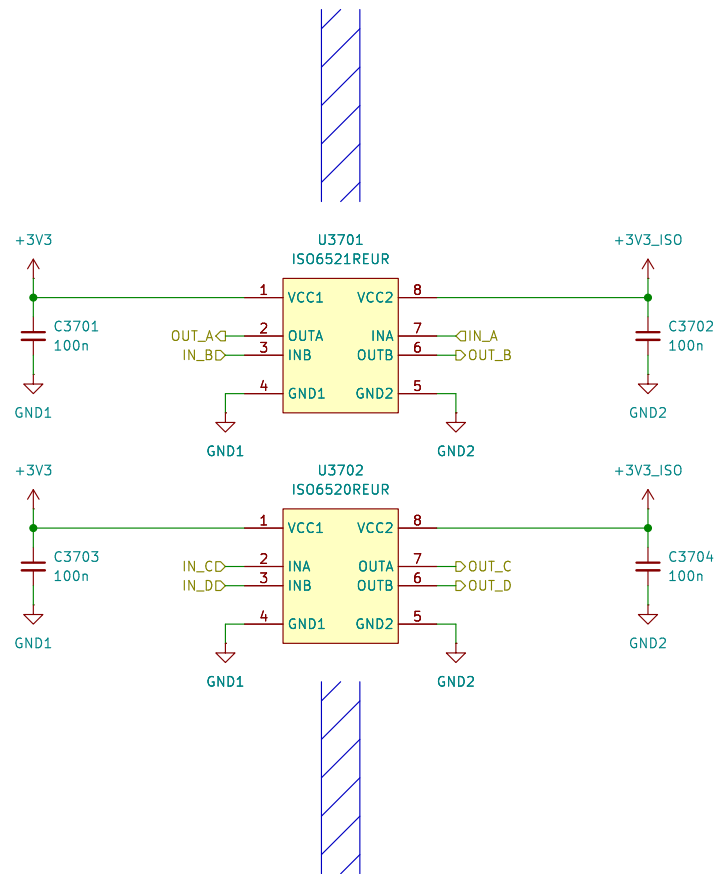
Title: 1000BASE-T1 CDN

Size: A4
KiCad E.D.A. 9.0.7

Date:

Rev:
Id: 35/38





Sheet: /Isolation Crossing/
File: isolation-crossing.kicad_sch

Title: Isolation Crossing

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 37/38

